



OUR OCEAN 2017 commitments

Contents

Marine pollution.....	2
Marine protected areas.....	13
Maritime security.....	26
Sustainable blue economy.....	32
Sustainable fisheries.....	45
Climate change.....	56

For more information about the 2017 edition of the #OurOcean conference go to www.ourocean2017.org.

Disclaimer: Our goal is to accurate information. If errors are brought to our attention, we will try to correct them. However, the European Commission and the European External Action Service accept no responsibility or liability whatsoever with regard to this information. The announcement of commitments at the Our Ocean Conference and their inclusion in the final list of commitments does not mean in any way that the European Commission has endorsed their legality.



Marine pollution

As part of its upcoming plastics strategy, the **European Union** announced draft measures to reduce the leakage of plastics into the environment by the end of 2017.

The **European Union** announced draft measures in 2017 to reduce the discharges of ship-generated waste and cargo residues into the sea.

The **European Union** announced the launch of WISE-Marine, a gateway to information on European water issues for the general public and stakeholders to promote better ocean governance and ecosystem-based management. The platform will be expanded and integrated further in the years to come.

The **European Commission** announced that it will phase out by end 2017 all single-use plastic cups in water fountains and vending machines in all its buildings and meetings. It also committed to report on all its efforts towards a further reduction of the use of other single-use plastic items in all its building and events at the occasion of the 2018 Our Ocean Conference. Measures to achieve this will include improving its green public procurement, reducing single-use plastics in canteens and cafeterias, promoting use of tap water, launching a wider awareness raising campaign for staff on waste reduction, sorting and recycling and greening Commission events.

The **European Union** announced that it will contribute EUR 2 million in 2017 to support the implementation of the Marine Strategy Framework Directive by the Member States and a further 2.3 million to support regional and inter-regional cooperation for this objective. The EU law aims to achieve Good Environmental Status (GES) of the waters of EU Member States by 2020 and to protect the resource base upon which marine-related economic and social activities depend.

The **European Union** announced that it will commit EUR 2.85 million to marine pollution prevention and preparedness projects and EUR 2.5 million to marine pollution exercises, to support and complement the cross-border cooperation efforts between EU countries and with selected countries in the EU's vicinity.

Malta announced the establishment of a new government body, with the remit of studying, protecting, preserving, managing and educating on the country's rich underwater cultural heritage with a budget of EUR 1.6 million over 3 years.

Malta announced the launch of a the Small States Centre of Excellence that will provide practical support to government administrations of small states to attain the Sustainable Development Goals under Agenda 2030. The Centre will provide relevant knowledge based tools, practical solutions and share best practices to leverage a collective synergy, catalysing the development of small states in the context of their individual limited resources. This will be done primarily through the use of the Internet and online based services. The website will be designed around the universally-accepted UN SDG Knowledge Platforms, particularly SDG goal 14.1 and 14.2 (marine pollution and marine protected areas).

Malta announced to designate 30% of its waters as Marine Protected Areas in 2018 to ensure protection of caves and reefs. As part of this objective, Malta announced it would develop management plans by 2020 for the fourteen marine protected area sites it designated in 2016 comprising roughly 3,487 square kilometres, covering an area significantly larger than the country itself.

Malta announced it will commit to introduce a beverage container refund scheme by the end of 2019 to ensure that at least 70% of the plastic bottles generated in its islands are recovered and mitigate the impact of marine litter on the ocean.

Malta announced it would commemorate the pioneering role of the country in the creation of the UN Convention on the International Law of the Sea. The process within the United Nations General Assembly started in 1967 with the well-known speech of the Maltese Ambassador Dr Arvid Pardo. Celebrations will include a national symposium dedicated to the initiative by Dr Pardo in 2017, a side event at the UN on ocean governance in 2018 and annual lectures and promotional material (including dissemination of clips and documentaries shown on TV on a yearly basis) to underscore the relevance of his work and competitions on the Law of the Sea aimed at children.

HRH the Prince of Wales announced a blue economy initiative to facilitate collaborative efforts, ensure blue economy builds resilience; encourage ocean-related investments and policies to support sustainable development.

The **Lonely Whale Foundation** announced its commitment to permanently eliminating 15 billion single use plastic straws by 2020.

Sweden announced the allocation of EUR 15.6 million (SEK 150 million) for at least the next three years for the "Eco Bonus" initiative, to support a less harmful shipping with the aim of shifting freight from land to sea.

Sweden announced the allocation of EUR 53 million (SEK 528 million) in 2017 to address marine pollution in three main areas: removal of hazardous substances particularly in wrecks, treatment of the accumulated fiber banks from industrial process water and reduction of the amount of pharmaceuticals to the marine environment; combat eutrophication through for example the application of measures to reduce the internal load of phosphorus in lakes and coastal bays, the restoration and construction of wetlands, the development of blue catch-crops and the reduction of nutrients' outflow; management of plastic in a responsible way through actions including beach cleaning, development of new materials, cleaning of storm water and global support for biodiversity

Sweden announced the allocation of EUR 3.7 million (SEK 36 million) over the next three four years for a set of initiatives by the International Union for Conservation of Nature (IUCN) to support governments, industry and society in Eastern and Southern Africa and the Asia Pacific regions in promoting, enacting and enforcing legislation and other measures to contain and reduce marine plastic pollution

Sweden announced the allocation of EUR 10 million (SEK 100 million) over the next five years into the Pacific-European Union Marine Partnership Programme (PEUMP). The aim of the Programme is to support 15 members of the African, Caribbean and Pacific (PACP) states in improving the sustainable management and development of fisheries for food security and economic growth, while addressing climate change and conservation of marine biodiversity

Sweden announced the allocation of EUR 6 million (SEK 60 million) in 2017 to address to increase the efforts to establish an ecologically representative, well connected and effectively managed network of MPAs in Swedish waters. The commitment also includes efforts to reduce the bycatch of ghost gear.

Germany announced the launch of the G20 Action Plan on Marine Litter as part of its 2017 G20-Agenda. Upon proposal by Germany, the G20 Heads of State and Government have confirmed their commitment to protect the marine environment by adopting measures to address pollution from land-based and sea-based sources; provide financial resources for cost-effectiveness analyses, and to prevent or reduce marine litter. The plan sets effective actions, e.g. to facilitate the implementation of the polluter pays approach, 'extended producer responsibility' or deposit schemes; develop new sources of funding for effective waste management systems and to stimulate innovations. It further addresses education and awareness raising, as well as additional research needs. As part of the action plan, a voluntary "Global Network of the Committed" has been launched in order to support the implementation of measures defined in the action plan.

Germany announced to reinforce its continuous support for the EBSA process by dedicating EUR 100,000 to promote the designation of ecologically or biologically significant marine areas (EBSAs) within and beyond national jurisdiction under the Convention for Biological Biodiversity (CBD). By the end of 2017, Germany will organize an international workshop to advance the description of new areas, and for strengthening the scientific credibility and transparency of the EBSA process.

Germany commits EUR 30 million in 2017 to the implementation of development projects under the "Ten-point Plan of Action for Marine Conservation and Sustainable Fisheries". This includes inter alia funding for projects which serve the implementation of the G20 Action Plan on Marine Litter. Also in this context, Germany commits to further advance the implementation of the "Blue Action Fund" for marine conservation together with the Swedish Government that is joining the Blue Action Fund in 2017.

Germany announced the launch of the STRONG High Seas Project this year. Through the support by Germany and contributions from partners, the project will receive EUR 3.9 million over the next 5-years to strengthen regional ocean governance for the conservation and sustainable use of marine biodiversity in areas beyond national jurisdiction (ABNJ). The Institute for Advanced Sustainability Studies (IASS) will lead the project and together with the Institute for Sustainable Development and International Relations (IDDRI), BirdLife International, the International Ocean Institute (IOI) -Southern Africa, the Universidad Católica del Norte (UCN), WWF Colombia, and WWF Germany identify best practices and provide support to regional institutions and national authorities. Working with the Secretariats of the Comisión Permanente del Pacífico Sur (CPPS) and the Abidjan Convention, and others

it will develop and propose measures to support integrated and ecosystem-based ocean governance in the Southeast Pacific and Southeast Atlantic.

China announced EUR 8 million (USD 9 million) worth national research project to develop marine micro plastics, jellyfish, red tide, pathogenic microorganisms monitoring and preventing technology. China also announced its intention to focus on the elimination of plastic waste pollution in estuaries and bays and to formulate plans for action against pollution from marine wastes.

China announced that by 2020 it will implement the "Blue Bay" renovation project in at least 66 bays and adjacent areas, approximately 70% of the coastal water quality meets the excellent or good Grade (I or II Grade) for Chinese National Sea Water Quality Standard.

China announced that it will further strengthen the conservation of the natural coastline, in order to ensuring that the mainland natural shoreline remains at least 35 percent.

P&G Dish Care announced that will introduce a limited-edition Beach Plastic bottle, with 10 per cent of the resin bottle coming from recovered beach plastic and 90 per cent standard PCR. In partnership with recycling experts at TerraCycle, this innovation will come to the UK in 2018 available at one world's leading retailer. This action is the result of a strategic alliance between Terracycle and will serve to find innovative uses for beach plastic while raising consumer awareness of the ocean plastic issue. This initiative complements P&G's support of the Trash Free Seas Alliance efforts to dramatically reduce the flow of plastic into the world's oceans.

In an effort to send a strong demand signal for recycled resin, **P&G Dish brands** - the world's #1 selling handwashing liquid - announced it will continue to use 8,000 metric tonnes of recycled plastic per year in its transparent plastic bottles, using an average of 40% Post-Consumer Recycled plastic content across 481 million of our transparent dish care bottles globally. Given the size and scale of P&G Dish Care brands, these efforts will create a demand signal that will help support greater recovery of plastics, while enabling consumers to access recyclable and recycled products every day, without any compromises.

The **Plastic Solutions Fund** announced its intention to raise EUR 12.6 million (USD 15 million) over the next three years to combat the problem of single use and other disposable plastic packaging, with EUR 9 million already committed. The Plastic Solutions Fund has been established to support non-profit organizations, particularly in Asia, the European Union and the United States, that want to transform plastics packaging supply chains, reducing not only the amount of this plastic that enters the environment, but the amount that is manufactured in the first place. The Plastic Solutions Fund announced it has given EUR 2.1 million (USD 2.5million) in grants to fifteen organizations since January, ranging in size from EUR 21,000 (USD 25,000) to EUR 421,000 (USD 500,000) per annum.

France announced that it is committed to ban single-use plastic cups, glasses and plates by 1 January 2020. France also announced that it will no longer sell cosmetic products containing plastic microbeads by 1 January 2018 and cotton buds with plastic stems by 1 January 2020.

France announced EUR 500,000 to UNEP activities under the Global Partnership on Marine Litter (GPML) from 2017 to 2020.

France announced that it will create by 1st January 2018 a structured network that will address the issue of end-of-life recreational craft. By 2021, 26,000 ships will be dismantled. This action will help improve waste disposal and promote circular economy.

France announced that over the next five years it will develop cross-border cooperation under the Regional Marine Pollution Emergency Response Centre for the Mediterranean Sea (REMPEC) to prosecute polluters responsible for oil spills in the Western Mediterranean Sea.

France reaffirmed the second expansion of the marine reserve in the French Southern and Antarctic Lands in the Indian Ocean by 1 million square kilometers by 2022, which already includes a buffer zone. This will enable France to reach a total of 32% of French waters classified as Marine Protected Areas by 2022. France also reaffirmed that it will proceed with all stakeholders to manage 22% of its MPAs already established.

New Zealand announced it will ban all 'wash-off' products that contain plastic microbeads for exfoliating, cleansing or abrasive cleaning purposes. As well as personal care products, this includes household, car or other cleaning products. The ban will come into effect by May 2018.

New Zealand announced that it has joined the International Alliance to Combat Ocean Acidification.

Panama announced that by 2018 it will contribute more than EUR 850,256,000 [1 billion dollars] to the panama city and bay sanitation project to ensure the marine contamination is reduced and the health of Panama bay and marine ecosystems are restored".

Panama announced the ratification of the International Convention for the Control and Management of Ships' Ballast Water and Sediments ships 2004 (BWM) in Law 41 of September, 2016

Sri Lanka announced its commitment to make the country polythene-free and to find a sustainable solution to ocean pollution and solid waste management. To this end, it will take measures including the ban on the import, manufacture and sale of harmful polythene products and the ban on the use of polythene for decorations.

Sri Lanka also announced that, in 2017, it commenced projects aimed at encouraging the separation of waste with the aim of recycling and is also exploring sewage energy generation projects.

The **United States** announced a public-private partnership that has deployed a joint U.S.-EU public diplomacy and education exhibit on marine debris and plastics. A partnership with the EU, several U.S. Embassies, civil society, and multiple private sector organizations, the exhibit highlighted U.S. and EU leadership on ocean issues, including the Trash Free Seas Alliance. The exhibit traveled across the EU, from Copenhagen to Tallinn, Lisbon, and

Valletta, and later Genoa, reaching hundreds of thousands of visitors as a flagship joint U.S.-EU diplomacy effort on marine debris. For more detail, see <https://ourocean2017.org/campaigns/tackling-plastic-pollution-ocean>.

The **United States** announced a contribution of up to EUR 418,000 (USD 500,000) to the New Plastics Economy Initiative's Circular Design Challenge. A partnership among USAID, the Ellen MacArthur Foundation, and a coalition of private companies and foundations, the Challenge will identify solutions to advance the development of new packaging formats and alternative delivery models for plastics.

The **United States** announced up to EUR 7.6 million (USD 9.1 million) over four years for the establishment of a small grants portfolio on waste recycling that will focus on municipal waste management and recycling efforts in Indonesia, the Philippines, Sri Lanka, and Vietnam. The objective is to improve waste management practices and reduce the amount of marine-bound plastics. The program will also evaluate the effectiveness of promising grants and provide recommendations on how USAID may scale these efforts.

Unilever announced its commitment to help transform global plastic packaging material flows by: ensuring all of its plastic packaging is reusable, recyclable or compostable by 2025; increasing its use of recycled plastic content in its packaging to at least 25% by 2025 (against a 2015 baseline); publishing the full “palette” of plastics materials used in its packaging by 2020 to help create a plastics protocol for the industry; and helping tackle the industry-wide sachet waste issue, by investing in proving, and then sharing with industry, a technical solution to recycle multi-layered sachets - CreaSolv®.

The **Aquarium Conservation Partnership (ACP)**, an initiative designed to increase the collective impact of aquariums on shared ocean and freshwater conservation goals, announced the elimination of plastic straws and single-use take-away plastic bags in its 19 member aquariums as of 10 July 2017, and the significant reduction or elimination of single-use plastic beverage bottles in member aquariums by 1 December 2020. Founded by Monterey Bay Aquarium, Shedd Aquarium, and National Aquarium, the ACP also committed to showcasing innovative alternatives to single-use plastic for its millions of visitors. The members of the Aquarium Conservation Partnership are committed to allocate annually EUR 210,000.

The **National Aquarium** announced its commitment to divert at least one million single-use plastic bottles from the waste stream and ultimately the ocean by 2020, while also continuing not to offer single-use plastic bags or plastic straws or any plastic cutlery, and decrease by 50 per cent plastic throughout its operations by 2020.

Borealis AG announced an investment of EUR 15 million to advance mechanical recycling of Polyolefins, further to the acquisition of Germany's polyolefin recycling company mtm plastics GmbH.

Borealis AG announced a EUR 4 million initiative to accelerate waste management improvements in South-East Asia to be rolled out in 2018-2019, driven and to be co-funded with local and global partners.

Borealis AG announced its commitment to zero pellet loss from its operations and to substantially invest in best available technology to prevent pellet loss.

The **Dow Chemical Company** announced EUR 128,100 (USD 151,000) funding for three new research project to help solve the issue of plastics marine litter. Two projects, one each in Japan and Indonesia, will set up systems to measure the flow of trash into the ocean, and will then propose solutions, based on that information, to prevent the characterized waste from reaching the ocean. The third project will test the use of non-recycled plastic in roadways in Indonesia, to help create a new end-use market for collected materials that might otherwise end up in the environment. This will help drive collection of low value plastics, keeping these materials out of the ocean.

ECOALF, by means of its Foundation supported by HAP FOUNDATION, in partnership with ECOEMBES and with the collaboration of the Spanish Fishing Confederation, announced that it will be extending the "Upcycling the Oceans" project to the whole Spanish Mediterranean coast and test it also in the Atlantic Ocean. By 2018, the number of ports involved will be expanded from 32 to 60 and from 440 trawlers to 770, removing 200 tonnes of marine debris. The Project is helping rid the oceans of rubbish through partnership with the fishing sector and is giving a new life to the collected trash turning it into polyester garments, through a circular economy project.

The **GEF** announced a EUR 35.9 million (USD 42.3 million) commitment to the MedProgramme to address nutrient pollution in the Mediterranean, enhance region wide environmental monitoring and protect key coastal freshwater resources. The Mediterranean Sea Programme (the MedProgramme): builds on several successful GEF projects in the Mediterranean and on the legal framework provided by the Barcelona Convention and its protocols. The Programme is implemented by the United Nations Environment and the European Bank for Reconstruction and Development (EBRD), and will increase capacity across the region and result in a reduction of nutrients and chemicals reaching the Mediterranean Sea, which is currently causing coastal ecosystem degradation and potentially affecting the health and quality of life of millions of people in the coastal areas of the Mediterranean. This program consists of several subprojects, including those managed by EIB under the Mediterranean Hotspots Investment Programme (MeHSIP). The MeHSIP aims at assisting promoters with the preparation of projects in the environmental sector that address the main sources of pollution entering the Mediterranean Sea. A USD 7 million GEF contribution will allow MeHSIP to prepare subprojects leveraging an expected implementation value of USD 500 million.

Legambiente ONLUS announced its commitment to promote a ban in the Mediterranean countries on plastic carrier bags below 100 microns (except bags for primary transport) biodegradable and compostable certified EN 13432 or ISO 14855 through at least 30 events, within the campaigns Clean up the Med/spiagge e Fondali Puliti, involving other Mediterranean organizations.

Marks & Spencer announced that all its plastic packaging in the UK will be 100% recyclable and widely recycled by 2022. It will work to eliminate packaging that strays into the environment (particularly oceans) and actively design out packaging parts that can't be reused

or recycled. It will introduce products with reclaimed social plastics as a component, providing positive social benefit to the communities from which the materials are sourced. In addition, M&S will assess the feasibility for all its plastic packaging to be made from one polymer group by 2022 to reduce consumer confusion and to improve recycling.

Ocean Conservancy, the **Trash Free Seas Alliance**, and **Closed Loop Partners**, with the support of world leading brands, including **Procter & Gamble**, **3M**, **PepsiCo**, and plastic makers from the **American Chemistry Council** and the **World Plastics Council**, announced the formation of a new initiative to reduce marine debris through improving waste collection, recycling and reuse. The initiative, which will be led by Closed Loop Partners, aims to create a new funding mechanism to raise over EUR 128 million (USD 150 million) over five years to improve waste collection, sorting, and recycling markets in Southeast Asia. The funding mechanism will: catalyse new investments from the private sector, governments and development finance institutions; demonstrate solutions; build a pipeline of bankable waste management infrastructure projects to demonstrate investment viability; and maximize recycling profitability.

PepsiCo announced its ambition to design 100% of its packaging to be recyclable, compostable or biodegradable, increase recycled materials in its plastic packaging, reduce packaging's carbon impact, and in partnership with the PepsiCo Foundation, work to increase recycling rates by 2025.

Think Beyond Plastic announced the launch, in partnership with UN Environment, of the world's first student competition for solutions to the global problem of marine plastics (the 2017 Marine Plastics Innovation Challenge) which invites all university students to submit ideas in the fields of engineering, communications, economics and data modelling. Competition is accepting applications until November 20, 2017. Winners, which will be announced at Sixth International Marine Debris Conference, will gain entry into the Think Beyond Plastic annual acceleration program, which will provide mentoring and bridge to commercialization for winning ideas.

Think Beyond Plastic announced a partnership with California State University at Monterey Bay to develop an Innovation Center with a budget EUR 4.25 million (USD 5 million) to accelerate commercialisation of research and innovation with a focus on plastic pollution. This hub will bring together innovators, entrepreneurs, scientists, engineers, businesses, investors and consumer advocates to facilitate a multidisciplinary approach to solving this complex problem. This center will offer growing access to material innovation space for development, characterisation and scalability testing by 2020.

Volvo Ocean Race announced that in its 2017-2018 edition single-use plastic consumption in Race Villages will be reduced by at least 80%, with the objective to ban it entirely from 2019-2020 edition. In addition, Volvo Ocean Race will raise awareness on ocean plastic pollution by advocating "Turn the Tide" message to 3 million visitors to its Host Cities globally; by promoting UN Environment's "Turn the Tide on Plastic" campaign to its digital audience, including 1.2 million Facebook fans; by launching a dedicated "Turn the Tide on Plastic" sailing team with a powerful call to action via the Clean Seas programme; by using compelling storytelling and the race's significant media power to amplify the campaign - with

a minimum goal of 30% of external media articles mentioning its sustainability programme; and by developing an Education Programme for kids and teachers, with an initial delivery target of 25,000 students at Host Cities globally and online. Volvo Ocean Race will host 7 "Ocean Summits" by the June 2018, gathering more than 2,000 decision makers and influencers from business, government and science to make commitments to ocean health. Finally, Volvo Ocean race will contribute in 2017-2018 to ocean science by gathering data and monitoring micro plastic pollution on-board race boats.

The **Werner & Mertz Group**, whose packaging is already hundred per cent recyclable, announced its commitment to use hundred per cent recycled plastic in at least 70 million bottles each year as of 2017, corresponding to sixty-five per cent of its entire annual bottle volume, aiming to go up to one hundred per cent for all its consumer goods packaging by 2025. The **Werner & Mertz Group**, whose packaging is already hundred per cent recyclable, announced its commitment to use hundred per cent recycled plastic in at least 70 million bottles each year as of 2017, corresponding to sixty-five per cent of its entire annual bottle volume, aiming to go up to one hundred per cent for all its consumer goods packaging by 2025. The **Werner & Mertz Group**, whose packaging is already hundred per cent recyclable, announced its commitment to use hundred per cent recycled plastic in at least 70 million bottles each year as of 2017, corresponding to sixty-five per cent of its entire annual bottle volume, aiming to go up to one hundred per cent for all its consumer goods packaging by 2025.

Zero Waste Europe, a European network of NGOs working with 397 zero waste municipalities, announced the allocation of EUR 300,000 over the next three years to include other 20 cities into the network, thus reducing waste generation by 20% and increasing recycling by 40%.

Zero Waste Europe, in collaboration with the Global Alliance for Incinerator Alternatives in the Philippines and 9 other partners in South-East Asia, announced the allocation of EUR 300,000 to involve 16 cities in South East Asia to implement a zero waste strategy by 2020, preventing more than 868,000 tons of annual waste from entering the environment and including more than 173,000 tons annually from being released into the environment.

The **United Kingdom** announced EUR 566,700 (GBP 500,000) to launch a new 'litter innovation fund' to trial small scale projects that could be replicated more widely, including those aimed at reducing litter entering the marine environment.

The **United Kingdom** announced the adoption of a Litter Strategy for England. The strategy aims to apply best practice in education, enforcement and infrastructure to deliver a substantial reduction in litter and littering behaviour on land, which in turn will lead to a reduction in the amount of litter reaching the marine environment. The Government will work with industry and community partners to implement this strategy. This is in addition to existing Litter Strategies in Scotland and Northern Ireland.

The **United Kingdom** announced its commitment to banning plastic microbeads in rinse-off cosmetics and personal care products to protect the marine environment from avoidable

microplastic pollution. It anticipates having bans on both manufacture and sale of such products in place across the whole of the UK by the end of June 2018.

The **Surfrider Foundation Europe** announced the launch of the #ResetyourHabit campaign. The initiative aims at creating awareness on the impact of single-use water plastic bottles and inspiring people to change their habits by using reusable gourds and canteens; industries to change the design of beverage bottles by preferring high-capacity and refillable containers; private companies and public institutions to use and promote tap water and water fountains.

The **Surfrider Foundation Europe**, with the support of the Territorial Council of Guipúzcoa, AZTI Tecnalia and Rivages Pro Tech Research Centers, the Syndicat mixte Kosta Garbia, and the municipality of Biarritz, announced the launch of the LIFE LEMA project. The initiative will provide methodological guidance and intelligent tools to local authorities for the effective management of floating marine litter with a cross-border approach whilst promoting the diversification of fishing vessels' economic activities. LIFE LEMA will run until 2019. It entails a total investment of EUR 2.1 million co-financed by the European Union.

Turkey announced that it will develop Marine Litter Action Plans for all coastal cities by the end of 2018. Clean-up activities, pollution reduction studies will be reported annually and minimization of marine litter will be conducted with relevant sectors such as plastics, cosmetics and textile within the framework of above mentioned Action Plans. Starting from 1 January 2019 pricing of plastic bags will be on the agenda.

The **Republic of Palau** announced that it has introduced a legislation to ban the importation of plastic bags, phasing out by 2025.

Ghana announced its commitment to eliminate pollution along the country's coast and significantly reduce pollution in the marine ecosystem by 2025, by tackling the current challenges posed by the use of plastics and indiscriminate disposal of waste.

Norway announced a multi-annual programme to assist developing countries in improving waste management to prevent land-based litter from ending up in the ocean, including plastics and microplastics. Beach and coastal clean-ups may also be part of the initiative. The programme will be launched in 2018 with NOK 150 million - approximately EUR 16.2 million - set aside for the first year.

Indonesia announced it will launch a National Action Plan on Marine Plastic Debris to achieve a reduction of 70% of its plastic debris by the end of 2025, therefore contributing to its national ambition to become trash-free. Indonesia also announced it will invest up to EUR 0.85 billion to develop a national programme to address the management of waste from land-based sources over the next four years. In addition, Indonesia announced the inclusion of the issue of Marine Plastic Debris in its national education programme.

Thailand announced the establishment of the Pollution Management Plan 2017-2021 to maintain the quality of the marine environment and tackle marine pollution from land-based activities. Thailand also announced the adoption of the National Master Plan on Waste Management 2016-2021 and will launch the National "3R" (Reduce, Reuse, Recycle) strategy

and a Plastic Debris Management Plan to reduce plastic waste by 2021. The plan consists of 3 key measures, namely 1) raising awareness of plastics usage and its effect on marine environment, 2) reducing plastics usage, and 3) enhancing research capacity in field of plastic material/plastic substitute and marine debris assessment. The first campaign will be launched in order to end the use of drinking water bottle cap-seal which will help reduce 520 tons of plastic waste annually.



Marine protected areas

The **European Union** announced that it will commit EUR 20 million to support the management of marine protected areas in African, Caribbean and Pacific countries through the programme BIOPAMA II (Biodiversity and Protected Areas Management Programme).

The **European Union** and **Germany** announced that they will support the establishment of a cross-sectoral and cross-boundary multi-stakeholder platform for regional ocean governance by 2020. This platform will be developed under the Partnership for Regional Ocean Governance (PROG), initiated in 2015 by the United Nations Environment Programme (UNEP), the Institute for Advanced Sustainability Studies (IASS), the Institute for Sustainable Development and International Relations (Institut du Développement Durable et des Relations Internationales - IDDRI) and the Think Tank for Sustainability (TMG). The development of the platform has been announced by Germany as a voluntary commitment on the occasion of the UN Ocean Conference for their implementation of SDG14 (5-9 June 2017). The PROG forum will provide new knowledge on integrated ocean governance at three different levels: (1) within regions; (2) between regions; and (3) between the regional level and the global level. Building on a collaborative process with international partners in 2018, the European Union and Germany will organise the first meeting in 2019.

The **European Union** announced that it will invest EUR 1.5 million to analyse ecosystems and economic activity on the mid-Atlantic Ridge and the Rio Grande Rise, in order to support the definition of a coherent set of Areas of Particular Environmental Interest.

The **European Union** announced its intention to support the General Fisheries Commission for the Mediterranean in establishing a Fishing Restricted Area (FRA) of at least 2,700 square kilometres to protect demersal stocks in the habitat recognised as essential nursery and spawning ground for a number of marine species outside territorial waters of Italy and Croatia of the Jabuka/Pomo Pit area of the Adriatic Sea. The creation of the Jabuka/Pomo Pit FRA will be for decision at the annual session of the General Fisheries Commission for the Mediterranean (GFCM) on 16-19 October 2017.

H.S.H. Prince Albert II of Monaco announced a commitment of EUR 250,000 to the "Mediterranean MPA Trust Fund" and EUR 200,000 to the "Beyond Plastic Med Initiative" (BeMed) in 2018.

The **Ellen MacArthur Foundation** announced and awarded the winners of its Circular Design Challenge with EUR 850,000 (USD 1 million) prize money. The challenge was launched together with the Prince of Wales's International Sustainability Unit and is funded by Wendy Schmidt. Together with a 12-month accelerator programme, the prize money will help scale design solutions that keep plastics in the economy, and out of the ocean. The challenge is part of the Ellen MacArthur Foundation's three-year New Plastics Economy initiative, applying the principles of the circular economy to build a plastics system that works.

Chile announced the creation of three new Marine Protected Areas - Juan Fernandez Archipelago, "Seno Almirantazgo" (Tierra del Fuego) and Cabo de Hornos and Diego Ramirez Archipelagos - which will add 620,152 square kilometres to the existing MPAs. With these three new areas almost 46 percent of Chile EEZ will be under protection.

Chile reaffirmed that by December 2017 the proposal for a National Ocean Policy will be presented to the President. This National Ocean Policy aims to ensure effective coordination among public institutions responsible for all the actions that take place in the conservation and sustainable use of the ocean.

Chile announced the creation of National Committee to Combat Illegal, Unreported and Unregulated Fishing (IUU). This Committee aims to strengthen and coordinate Chile's responsibilities as a coastal, port, flag and market State to combat illegal fishing activities and improve the co-ordination among relevant stakeholders for the development of mid-term action plans. This Committee is chaired by the Minister of Foreign Affairs and includes all public institutions related to the fishing industry, coastal communities and ocean activities.

Chile reaffirmed its commitment to sustainable fisheries and announced that as of August 2017 it has become a Cooperating Non-Contracting Party of the Inter-American Tropical Tuna Commission (IATTC).

Chile announced that it will ban plastic bags in coastal cities throughout the country. Once the law is approved by the National Congress, Chile will be the first American country to implement this type of ban.

The **Cook Islands** announced to become 100% reliant on renewable energy by 2020. They have already achieved their target of 50% since 2015.

The **Cook Islands** announced the achievement of their 2014 commitment that was to legally designate a multiple-use marine park over their entire Exclusive Economic Zone of 1.9 million square kilometres including exclusion zones where no large scale commercial fishing or seabed minerals activities will be permitted. With the passing of the Marae Moana Act 2017 there are now 15 exclusion zones encircling each of the 15 islands of the Cook Islands to a distance of 50 nautical miles from shore. The total area of the exclusion zones is 324,000 square kilometres.

Canada reaffirmed the identification in May 2017 of a portion of Canada's Offshore Pacific Bioregion as an area of interest (AOI) for possible future protection as an Oceans Act Marine Protected Area (MPA). The Offshore Pacific AOI is located in the southern portion of the Offshore Pacific Bioregion extending from the continental shelf break, west of Vancouver Island, to the Exclusive Economic Zone (EEZ) boundary with an area of approximately 140,000 square kilometres. The interim conservation objective of the Offshore Pacific AOI is to contribute to the protection and conservation of the unique seafloor features (i.e., seamounts and hydrothermal vents) and the ecosystems they support in Canada's Offshore Pacific Bioregion.

Canada reaffirmed designation of 26,600 square kilometres, an additional 0.46% of marine territory in Canada as other effective area based conservation measures in June 2017. These existing fisheries management measures in Canadian waters provide refuge to fish, mammals and their habitat. They have been found to meet the five criteria, which Canada has developed, that conservation measure must meet in order to contribute to Canada's marine conservation targets: the geographic location must be clearly defined; stock management or conservation objectives need to directly relate to an important species or habitat; the area must contain an important species and important habitat; the measures must be long term; and, the measures need to protect the important species and its habitat from both existing and foreseeable pressures.

Canada reaffirmed the designation in June 2017 of St. Anns Bank Oceans Act Marine Protected Area, which provides protection from the impacts of human activities, adding an additional 4,364 square kilometres of protection or 0.08% of Canada's oceans. Located to the east of Cape Breton, Nova Scotia, St. Anns Bank contains ecologically and biologically significant features including unique habitats, areas of high biodiversity, and areas of high biological productivity.

Canada announced the introduction in June 2017 of regulations to finalize the Laurentian Channel Marine Protected Area. The Laurentian Channel contains sensitive benthic habitats and is described as having one of the highest sea pen concentrations within the entire Newfoundland and Labrador Shelves Bioregion. The proposed Regulations would establish two management zones that provide varying levels of protection within the designated area, covering 11,619 square kilometres or 0.2% of Canadian waters.

Canada announced the Miramichi Bay Closure of September 2017, a fisheries area closure off New Brunswick that aims to protect adult Atlantic salmon and an important migration corridor for the species. This population of Atlantic salmon has been in decline and is listed as a species of special concern. The fisheries area closure covers approximately 1,553 square kilometres or 0.03% of Canada's ocean area. In order to allow Atlantic salmon to use the migration habitat as a way to return each year to the Miramichi River to spawn, the use of gillnets for all commercial groundfish fisheries is prohibited within the closure area.

Canada announced the September 2017 designation of the Western/Emerald Banks Conservation Area as a restricted fisheries zone off the coast of Nova Scotia. It is approximately 10,234 square kilometres, covering 0.18% of Canada's ocean area. The conservation objectives of this area are to support productivity objectives for groundfish species of Aboriginal, commercial, and/or recreational importance, particularly NAFO Division 4VW haddock, and manage the disturbance of benthic habitat that supports juvenile and adult haddock and other groundfish species.

Canada reaffirmed that in its 2017 budget, the Government of Canada has made a financial commitment of EUR 8,175,916 (CAD 12 million) over four years to protect the Last Ice Area, the only Arctic region expected to retain summer sea ice until 2050. As the last permanent ice-covered region in the Arctic, the area will be essential to ice-dependent species such as polar bears, beluga, narwhal, seals, walrus and the Inuit communities that depend on them for food. In partnership with Inuit, Fisheries and Oceans Canada will work together with all

partners including the Parks Canada Agency and other federal departments, other governments, management organizations, northern communities, and scientists.

Canada reaffirmed that it will be hosting an international technical expert workshop on marine other effective area-based conservation measures, February 6-9, 2018 in Montreal, and provide funding of EUR 110,605 (USD 130,000). This technical expert workshop will consolidate scientific and technical information on approaches for assessing the contribution and effectiveness of marine protected areas and other effective area-based conservation measures, as well as provide scientific and technical advice on the definition and identification of other effective area-based conservation measures for marine and coastal areas. This advice will then be used by the UN Convention on Biological Diversity Secretariat to inform the development of voluntary guidance on other measures.

Canada reaffirmed that it will host a pan-Arctic MPA science workshop in the autumn of 2018, which is intended to convene MPA managers, Indigenous peoples, community representatives, and other MPA partners to share experiences in planning and developing Arctic MPA networks, including the role that Indigenous people may play in developing and managing protected areas and other measures. The Arctic Council's Framework for a Pan-Arctic Network of Marine Protected Areas sets out the vision for an ecologically connected, representative and effectively-managed network of protected and specially managed areas. Further technical work and coordination at the pan-Arctic level is needed to advance this vision. As a step towards contributing to the Aichi target of 10% marine protected areas, this is one of a series of MPA science workshops being organized by the project co-leads, during 2017-2018.

Canada reaffirmed that it joined the UN Environment Clean Seas Campaign in July 2017 as part of Canada's commitment to take action on marine litter. The Government of Canada will continue to work with provincial and territorial governments, under the auspices of the Canadian Council of Ministers of the Environment (CCME), to improve Canada's record on reducing and recycling waste and implement the CCME waste action plan.

Canada reaffirmed that in June 2017 it published the Microbeads in Toiletries Regulations that will prohibit the manufacture, import and sale of all toiletries that contain plastic microbeads including cosmetics. The regulations will prohibit the manufacture, import, and sale of toiletries used to exfoliate or cleanse that contain plastic microbeads, including non-prescription drugs and natural health products. For the purposes of the regulations, plastic microbeads include any plastic particle equal to or less than 5 mm in size. The types of toiletries covered include products such as bath and body products, skin cleansers and toothpaste. The ban on sale, manufacture and import for all toiletries and natural health products will be phased in from January 1, 2018 to July 1, 2019.

Canada reaffirmed EUR 772,624 (CAD 1,134,000) in support announced in February 2017 for two new research projects to monitor contaminants and investigate their impacts in the Pacific and Arctic Oceans, in partnership with the Vancouver Aquarium. Fisheries and Oceans Canada is providing EUR 271,850 (CAD 399,000) to the Vancouver Aquarium to help implement Pollution Tracker. This project will sample mussels and near-shore sediment along the coast of British Columbia to collect data on a wide range of contaminants. A further

EUR 146,485 (CAD 215,000)is being provided for the Vancouver Aquarium study, for the first time, microplastics in the Arctic Ocean and their biological effects on marine life. An additional EUR 354,290 (CAD 520,000)in in-kind support, such as vessel use, will be provided by Fisheries and Oceans Canada to assist in the collection of samples.

Canada reaffirmed that it will host the 5th meeting of scientific experts on Fish Stocks in the Central Arctic Ocean in Ottawa, Canada on October 24-26, 2017 with the aim of finalizing a draft framework for implementation of a Joint Program of Scientific Research and Monitoring for the Central Arctic Ocean. Implementing this Joint Program will improve the state of knowledge and provide the baseline data needed to support informed decision making regarding any future fisheries management measures for the area.

Canada reaffirmed the establishment of the Coastal Restoration Fund to address historically degraded areas and will support projects that contribute to coastal restoration plans; support the identification of restoration priorities; and threats to marine species located on Canadas coasts. The Government of Canada will make EUR 51,099,473 (CAD 75 million) available from 2017-2022 for eligible projects.

Canada announced it has recommended boundaries for the Scott Islands marine National Wildlife Area. The Scott Islands make up one of the most productive and biologically diverse marine ecosystems, particularly for seabirds, on the Canadian Pacific Coast. The Scott Islands support the highest concentration of breeding seabirds in the Canadian Pacific, and are the site of the most intensive seabird research in Canada. The Scott Islands NWA encompasses approximately 11,546 square kilometres of an exclusively marine area that does not include the terrestrial portions of the Scott Islands, which are provincial protected areas.

Canada will continue to advance cooperation under the Canada-EU-United States Atlantic Ocean Research Alliance and further the successful implementation of the Galway Statement. Canada will continue to support the coordination and collection of data on the seabed through the hiring of a research vessel coordinator to identify trilateral at-sea opportunities for improved collaboration on vessels, and the North Atlantic trilateral survey transects such as those undertaken on board the CCG Louis St. Laurent (July 2015 and July 2016). We remain committed to our contributions to important trilateral science consortiums such as those associated with the Atlas and SponGES projects.

Niue announced that over the next three years 40 per cent of its Exclusive Economic Zone, equal to 127,000 square kilometers, will be declared as a large scale marine protected area. The area will include an offshore biodiverse rich Beveridge Reef (with the highest density of grey reef sharks in the world), a raft of resource rich deep ocean seamounts and the adjacent Territorial seas of Niue. This MPA will be established through the Niue Ocean Wide (NOW) project, a public-private partnership between the Government of Niue (GON) and Tofia Niue (a local non-profit organization) and key collaborating partners including Oceans 5, National Geographic Pristine Seas, and Niue Ridge to Reef project.

Sky, a major European media company, announced the launch of the Sky Ocean Rescue campaign in January 2017, aiming at raising awareness of how plastics and other pollution are

affecting the oceans. Through their news reports, documentaries, social media and a UK tour by the giant whale "Plasticus", Sky has engaged over 6 million people.

Sky announced the removal of all single-use plastics from its operations, products and supply chain by 2020.

Sky announced EUR 30 million over 5 years to create an Ocean Rescue Innovation Fund to develop ideas and technology to stop plastics entering the ocean.

Argentina announced its Government presented a draft law for the creation of two new marine protected areas. These areas are Yaganes and the extension of Namuncurá-Burdwood Bank, which will increase the coastal area to 155,918 square kilometres and constitute the first step to achieve the protection of 10% of these areas. The draft law proposes the creation of the Estricta National Marine Reserve and the Namuncurá-Burdwood Bank National Marine Reserve; and the Estricta National Marine Reserve, the Marine National Park and the Yaganes National Marine Reserve. These areas are located in the Exclusive Economic Zone of the Argentine Sea. Yaganes contributes 0.23% of the fishing effort at national level and will add more than 69,000 square kilometres to the coastal marine protection; while Burdwood represents 0.81% of the fishing activity with an area of more than 28,000 square kilometres.

Australia announced EUR 238,000 (USD 280,000) for "A New Song for Coastal Fisheries - Pathways to Change" (New Song), addressing coastal fisheries management, a new approach developed with Australia's DFAT funding in 2015. The main aim is to strengthen the regional community-based ecosystem approach to fisheries management (CEAFM) as outlined in the "New Song" strategy, while ensuring communities have a voice in this process at all levels. The project aims to empower and proactively engage communities through better representation and coordination of CEAFM assistance utilising mechanisms that facilitate a coordinated region-wide approach. The project will work closely with partners and NGOs to strengthen existing networks for gathering and sharing information on "who is doing what where", and then map this out in collaboration with the partners.

Australia announced the update of the threat abatement plan prepared in 2009, by mid-2018 aiming at providing national guidance on specific action to prevent and mitigate the impacts of marine debris. Injury and fatality to vertebrate marine life caused by harmful marine debris was listed as a key threatening process under Australia's Environment law. The plan update addresses six objectives, including the removal of existing marine debris and the increase of public awareness of the issue.

Australia announced EUR 10 million for the Integrated Marine Observing System (IMOS), launched in 2006 as a national collaborative research infrastructure. IMOS routinely operates a wide range of observing equipment throughout Australia's coastal and open oceans, making all of its data accessible to the users.

Australia announced EUR 24.7 million (USD 29 million) to the Climate and Oceans Support Program in the Pacific (COSPPac), supporting Pacific Island countries to adapt and mitigate the impacts of climate variability. COSPPac is successfully building capacity in Pacific Island countries National Meteorological and Hydrological Services and relevant Lands and Survey

Departments to develop and disseminate user-focused products and services that assist governments and communities to better prepare for severe climate events. The program has ensured that the Pacific regional partners are equipped for the transition and end of the current phase of COSPPac in June 2018.

Australia announced a series of initiatives within the International Partnership for Blue Carbon, including a workshop in the Philippines in August 2017 helping the Coral Triangle countries (Philippines, Malaysia, Indonesia, Solomon Islands, Timor Leste, PNG) identify needs and opportunities to develop tools supporting knowledge sharing. In addition, Australia is hosting visiting fellows from IORA countries (Madagascar, Mauritius and Seychelles) in November 2017 and supporting a proposed IORA summit in 2018.

Australia announced EUR 204,000 (USD 240,000) to create a bilingual web-based tool to improve fish identification and fisheries monitoring. Available in 2020, the tool will have a wider South-East Asian application. All fisheries monitoring programs require accurate fish identification and fisheries assessments routinely rely on fish catch data, as recorded by port-based enumerators and/or on-board observers.

Australia announced EUR 300,000 (AUD 447,000) for a series of events to foster greater blue carbon awareness and cooperation in the Indian Ocean region. As a first step, Australia will hold a symposium in 2018 to develop a common understanding of blue carbon and a framework for future priorities in the Indian Ocean region.

Australia announced EUR 2 million (AUD 3 million) for the Blue Economy Challenge and "Aquacelerator" accelerator program to support the aquaculture industry, develop local economies across the Indian Ocean region and connect inspiring innovators with the networks capable of turning their ideas into reality.

Australia announced EUR 2,8 million (USD 3.29 million) for a four-year investment to increase capacity in Pacific island countries (PICs) to prevent, deter and eliminate illegal, unreported and unregulated (IUU) fishing in the region. The Niue Treaty Subsidiary Agreement (NTSA) is a multilateral treaty to strengthen fisheries management and provide for more cost-effective and efficient maritime surveillance in the Pacific region. Currently the NTSA has ten parties and entered into force on 30 July 2014. This project will support PICs through the early technically-demanding stages of ratification and implementation of the NTSA by: (i) enhancing regional information sharing and cooperation; (ii) supporting risk-responsive tasking of assets; and (iii) increasing capacity to undertake enforcement operations.

Australia announced EUR 1.3 million (USD 1.6 million) for the Pacific Maritime Boundaries Project which aims to assist Pacific island countries (PICs) to secure maritime jurisdictional rights consistent with international law. Running from 2017 to 2020, the project will focus on two main areas: technical and legal support to the Pacific Community (SPC) and PICs to finalise outstanding Pacific maritime boundaries and limits; and establishing an administrative infrastructure and capability required for boundary dissemination and which also supports the efficient use of marine space and the resolution of conflicts between users.

Australia announced the building of a tool for the implementation of government policies by 2021 to guide and prioritise the Australian Government's activities to protect the environment. The Australian Government is committed to making ongoing improvements that enhance the currency and usability of this resource and will be exploring how it can make data available more frequently. Australia announced the release of the State of the Environment report as an interactive digital platform which enables decision-makers, researchers and the public to explore, compare, filter and download its findings in a variety of ways. This tool is an important resource for helping to guide and prioritise the Australian Government's activities to protect the environment, including the Marine environment. The Australian Government is committed to making ongoing improvements that enhance the usability of this resource and will be exploring how it can harness the value of making data available more frequently.

Australia announced that its national research agency, CSIRO, is leading a project, with a budget of EUR 1,333 million (AUD 2 million), from 2017-2020. Its objective is to use field sampling and mathematical modelling to document the distribution of plastic in the ocean on the coast and in the nearshore environment generated by 6-8 major urban centers and surrounding areas that have been identified as having significant waste mis-management or losses into the marine environment.

Cambodia announced the establishment of Marine Fisheries Management Areas which cover an area of 120 square kilometres in the Kep Province as well as its commitment to combat Illegal, Unreported, Unregulated fishing. Cambodia further reaffirmed its previous commitment to establish its first Marine Protected Area, which covers 405 square kilometres in the waters of the Koh Rong Archipelago.

Colombia announced the expansion of the fauna and flora Sanctuary of Malpelo by 17,090 square kilometres, increasing its extension from 9,500 to 26,670 square kilometres. With this decision, Colombia is protecting marine ecosystems with high influx of sharks, marine mammals and diversity of fish, whilst conserving the ecosystems associated with the ridges of Malpelo and Yurupari. With the recently declared marine protected areas in the Pacific, Colombia is exceeding the Aichi Target of the Convention on Biological Diversity and the national target for hectares of protected ecosystems, as embedded in the National Development Plan.

Costa Rica announced that on June 8 a memorandum of understanding was signed between the Tropical Eastern Pacific Marine Corridor (CMAR, a regional voluntary initiative by the governments of Colombia, Costa Rica, Ecuador and Panama) and PACIFICO (a coordination platform made up of four environmental funds). This strategic alliance seeks to mobilize and implement resources in common geographic areas in the Tropical Eastern Central Pacific in order to help consolidate the integrated management of ecosystems, biodiversity and marine and coastal resources through their conservation and sustainable use. To achieve this, the agreement already has a Five-Year Work Plan (2017-2022) and its first Annual Plan.

Croatia announced the expansion of Jabuka/Pomo Pit Marine Protected Area, the habitat recognised as essential nursery and spawning ground for a number of marine species. With this enlargement, MPAs cover more than 11 per cent of marine waters under national jurisdiction.

Croatia announced EUR 24 million over the five years to establish effective management framework for Natura 2000 network, including the preparation of management plans for at least 90 Natura 2000 marine sites. Croatia also announced the implementation of a five-year national capacity building program, starting in 2017, to train Natura 2000/protected area managers, including managers of marine protected areas.

Croatia announced EUR 13 million over five years for the preparation of detailed marine habitats mapping for waters under national jurisdiction, as the scientific basis for the determination of new marine protected areas in the Adriatic.

Croatia announced that by the end of 2018 it will set up a national monitoring system, including monitoring of marine species and habitats, to assess the effectiveness and efficiency of species and Natura 2000 sites management and will allocate EUR 10 million over five year period.

Croatia announced its commitment by 2020 to determine the most appropriate conservation areas for marine turtles and dolphins and in that regard has started research in 2017.

Croatia announced that during the next 5 years it will implement a management regime of marine resources, which will include spatial and temporal regulations within the inner and territorial waters over the area of no less than 20 percent of its territorial and inner waters.

The **Government of Myanmar** announced its commitment to develop a marine spatial plan by 2021; it should cover approximately 485,000 square kilometers of Myanmar's exclusive economic zone. The plan foresees developing further the Myanmar's marine protected area network, providing support to sustainable fisheries, as well as ensuring the development of a sustainable blue economy for all marine stakeholders.

Mexico announces the establishment of Revillagigedo National Park of 140,000 square kilometres, protecting the islands and surrounding waters, with a total ban of fishing activities in the area.

Romania reaffirmed its commitment to prepare a national plan to enhance institutional capacities and cooperation with maritime sectors in order to achieve good environmental status in the Black Sea marine region by 2020, notably the protection and preservation of the marine ecosystem.

Environmental Law Institute announced the development of a research and a policy dialogue on environmental displacement due to climate change in the Marshall Islands with grants totaling EUR 19,000 from the Ocean Policy Research Institute and the University of Tokyo.

Environmental Law Institute announced its continued effort under the ELI Gulf of Mexico Restoration and Recovery Initiative, through which ELI attorneys analyze the variety of restoration funding processes in the wake of the Deepwater Horizon oil spill, and translate them in accessible ways for local stakeholders. This project has been made possible thanks to a generous EUR 468,000 grant from the Walton Family Foundation.

Environmental Law Institute announced a research project on assessing legal options for the use of next generation sequencing and environmental DNA for conducting environmental impact assessments of open water aquaculture facilities and deep seabed mining operations, in cooperation with the University of Geneva. The project is developed with a EUR 50,000 grant from the Swiss Network for International Studies.

Environmental Law Institute announced a project to research ways to strengthen indigenous rights to fish and food sovereignty, as well as co-management models for fisheries security in Inuit communities in the Arctic, equipped with a EUR 158,000 grant from the National Science Foundation.

Environmental Law Institute announced that, during 2018 and 2019, it will develop a guide to help environmental impact assessment practitioners better estimate the economic and sociocultural impact of fisheries regulatory reforms on fishing communities in the United States. The project has been made possible thanks to a EUR 64,700 grant from the United States National Oceanic and Atmospheric Administration.

The **International Association of Oil & Gas Producers (IOGP)** announced that version 2 of its Seabed Survey Data Model (SSDM) was launched in June 2017. Designed in 2011 and built on decades of oil & gas industry surveys, the data model is now freely and widely available as a global standard for marine survey data, benefitting companies and organisations beyond the oil & gas sector. Available in open standard "geodatabase" format, the model includes provision for data such as bathymetric, seabed morphology, natural and man-made hazards, soil & environmental samples, and also shallow sub-seabed geological information.

The **International Association of Oil & Gas Producers (IOGP)** community and the UN Environment World Conservation Monitoring Centre announced that by 2017 9 leading oil and gas companies (IOGP members) are voluntarily participating in the Proteus Partnership initiated by 5 companies in 2003. The partnership supports biodiversity information development that becomes accessible to the wider conservation and scientific community. The Oil & Gas sector contribution supports the accuracy and completeness of information in the World Database on Protected Areas, supports the identification and assessment of marine biodiversity features and improves the availability of information on coastal and marine biodiversity to the benefit of the conservation and scientific community.

Marine Construction Technologies announced the reduction of underwater noise, resulting from impact-driving upon fish, marine mammals, and other sensitive wildlife, by 70-90 per cent per pile. It announced as well the reduction of construction-related noise pollution associated with steel marine foundations worldwide by 15 per cent by 2022.

The **MAVA Foundation** announced a EUR 70 million commitment over the next 5 years to the conservation of marine biodiversity mainly in the Mediterranean Basin and in West Africa for the benefit of people and nature. More specifically, MAVA Foundation will provide EUR 8 million to reduce environmental impacts of plastic pollution and oil and gas activities; EUR 5 million to stimulate a blue economy focusing on coastal infrastructure development; EUR 30 million to support sustainable fisheries for the benefit of local livelihoods and biodiversity;

and EUR 27 million to promote MPAs as a tool for protecting marine species, habitats and fish resources.

MedReAct, in collaboration with Stanford University, Marche's POLYTECHNIC University, Legambiente and Marevivo, announced the allocation of EUR 366,000 to launch the Adriatic Recovery Project. Running from 2017 to 2020, the initiative will help restoring essential fish habitats in the Adriatic Sea, by promoting through the General Fisheries Commission for the Mediterranean the establishment of Fisheries Restricted Areas closed to bottom trawling to protected key nurseries, spawning grounds and Vulnerable Marine Ecosystems in the region.

The **Nekton Oxford Deep Ocean Research Institute**, supported by **Kensington Tours**, announced a major global scientific collaboration from 2018 to 2021 that aims to create a step change in the knowledge of the Indian Ocean and catalyse its sustainable governance. The Mission consists of six research cruises in six distinct bioregions of the Indian Ocean from West (Mozambique Channel and Seychelles), to Central (Chagos and Maldives) and East (Andaman and Sumatra). The Mission will conclude with a State of the Indian Ocean Summit in November 2021. The Mission combines four major activities linked to clear measurable outcomes to deliver the greatest coordinated impact to accelerate the sustainable governance of the Indian Ocean: Scientific Research, Capacity Development Initiatives, Policy initiatives and Public Engagement and ocean literacy. The Mission will entail EUR 30 million (USD 35.4 million). 2. The Nekton Oxford Deep Ocean Research Institute announced a new framework for standardised multi-disciplinary marine research in conjunction with a dozen world leading marine scientists. From ocean physics to chemistry, from pelagic and benthic biology to geophysics, the new framework harnesses all major marine research technologies to support scientists to gather, analyse and publish comparable ocean data at scale and at speed. The Protocol aims to increase scientific productivity and the provision of actionable data that is required to catalyse ocean governance.

The **Nekton Oxford Deep Ocean Research Institute**, supported by **Kensington Tours**, announced a major global scientific collaboration from 2018 to 2021 that aims to create a step change in the knowledge of the Indian Ocean and catalyse its sustainable governance. The Mission consists of six research cruises in six distinct bioregions of the Indian Ocean from West (Mozambique Channel and Seychelles), to Central (Chagos and Maldives) and East (Andaman and Sumatra). The Mission will conclude with a State of the Indian Ocean Summit in November 2021. The Mission combines four major activities linked to clear measurable outcomes to deliver the greatest coordinated impact to accelerate the sustainable governance of the Indian Ocean: Scientific Research, Capacity Development Initiatives, Policy initiatives and Public Engagement and ocean literacy. The Mission will entail EUR 34 million (USD 40 million).

The **Tara Expeditions Foundation** announced under its Oceans program an investment of up to 9 million euros funded by scientific institutions and the French Facility for Global Environment. France has provided 2 million EUR to this initiative. The program will invest in research, innovation, transfer of technology and capacity building on plankton studies related to climate change. After collecting 35,000 samples and sequencing more than 40 billion genes, the Tara Expeditions Foundation announced that South American and African research institutions will join the program.

The **Tara Expeditions Foundation** announced under its Oceans program an investment of up to 9 million euros funded by scientific institutions and the French Facility for Global Environment. The program will invest in research, innovation, transfer of technology and capacity building on plankton studies related to climate change. After collecting 35,000 samples and sequencing more than 40 billion genes, the Tara Expeditions Foundation announced that southamerican and african research institutions joined the scientific program to improve analysis and share results to orient public policies.

Saildrone announced that by 2022 it will build a global ocean monitoring network consisting of 1,000 wind and solar powered saildrones providing planetary coverage at a 6x6 degree resolution, endeavouring to significantly reduce the cost of ocean data collection to improve management of sustainable fisheries, carbon cycle, marine pollution and climate change impacts on humanity. To further encourage scientific innovation, every year starting in 2018, Saildrone will award 30 days of saildrone-powered data, equivalent to approximately EUR 842,000 (USD 1 million) of research ship time, to the most impactful research project submitted.

The **Leibniz-Centre of Tropical Marine Research (ZMT)** reaffirmed its commitment to bring together scientists from 26 EU Member States to create and coordinate a research network for transdisciplinary research on ocean governance in the EU with the support of a grant of EUR 600,000 from the European COST programme. The network will identify ocean governance challenges in close cooperation with actors and stakeholders and provide integrated advice on governance tools and mechanisms to improve ocean related decision-making focusing on (a) land-sea interactions; (b) area-based management; (c) seabed resources; (d) nutrition security; (e) ocean acidification and climate change; and (f) fisheries governance.

Palau announced that it has met its 2020 goal of the Micronesia Challenge through the protection of 30% near shore marine resources.

Palau announced the creation of a Palau National Marine Sanctuary with the aim of working towards closing 80% of its EEZ to commercial fishing by 2020, making it a no-take. The remaining 20% is reserved for the fishing needs of the local population and tourism industry.

Palau announced that it will host 2020 Our Ocean Conference.

Ghana announced its commitment to complete the assessment of ecologically sensitive areas along the Ghanaian coast and the designation of Ghana's first marine protected area by 2025, to safeguard coastal and marine biodiversity.

Spain expressed its will to work on the expansion of the current maritime-terrestrial National Park of the archipelago of Cabrera and the establishment of the first exclusively marine national park in the Hierro Island, Canary Islands, with the consensus of regional and local authorities, and all sectors involved.

Thailand announced that it will establish 14,300 square kilometres of marine protected area (4.4 per cent of its EEZ) by 2020 as part of the implementation of its 5 years action plan on

area-based management and conservation measures. The proposed areas include 4 national marine parks (ca. 450 square kilometres), 3 environmental protection areas (570 square kilometres), and many marine and coastal protection areas, i.e. 25 island-groups (6,600 square kilometres) and one large offshore shelf zone (6,700 square kilometres). Thailand also announced its commitment to protect 10 percent of its coastal and marine protected areas by 2030.



Maritime security

The **European Union** announced that it will continue supporting maritime security in the Gulf of Guinea, including through the Gulf of Guinea Inter-Regional Network and the launch of two new programmes: the SWAIMS programme (Support to West Africa Integrated Maritime Security), worth EUR 29 million, and the programme to improving port security in West and Central Africa, worth EUR 8.5 million.

The **European Union** announced that it will provide EUR 1 million in 2017 to support the upgrading of the ICT systems of EU maritime authorities and facilitate cooperation between them. Furthermore, the European Union announced that it will contribute EUR 80,000 to facilitate cooperation between coastguard authorities in Europe.

The **European Union** announced to commit EUR 37.5 million to ensure maritime security and countering piracy along the south-eastern African coastline and in the Indian Ocean. The funds are to be implemented by four regional organisations (IGAD, COMESA, EAC and IOC) in cooperation with UNODC, INTERPOL and FAO. The programme supports alternative livelihood initiatives in the coastal pirate areas of Somalia, investigation capacities at national and regional level, prison reforms, prosecution and judicial capacity, disruption of illegal financial flows, combating money laundering, and various other maritime tasks, in addition to a regional mechanism for the coordination and exchange of maritime information.

The **European Union** announced the launch of a prototype surveillance tool in September 2017 which detects ships to reveal the extent of human activities at sea. The 'Search for Unidentified Maritime Objects' tool, or 'SUMO' for short, is a piece of software that automatically analyses data from radar imaging satellites to find vessels as small as 1 metre long, even in cloudy conditions or at night. The SUMO tool is open source, to promote uptake by users and developers and facilitate international cooperation on mapping of ship routes, monitoring shipping intensity, identifying polluting ships, monitoring fishing activities, countering piracy and smuggling, and controlling maritime borders.

The **European Union** announced that it will invest EUR 4 million in its satellite monitoring programme (Copernicus) in 2017 to support EU agencies and EU Member States in monitoring oil pollution and large-scale commercial fisheries (including the fight against illegal, unreported and unregulated fishing) in the Northwest Atlantic, the Mediterranean, the Baltic, the North Sea, the Black Sea, the Pacific Ocean and around the Canary Islands. Copernicus will also introduce new services to support law enforcement and navigation safety in ice-infested areas.

The **Seychelles** announced to develop a National Maritime Security Strategy by 2018.

The **United Nations Office on Drugs and Crime** announced that they will provide support through the Global Maritime Crime Programme to five member states covering West and East Africa, South Asia and the Caribbean in 2018 in conducting sea operations with the use of satellite imagery to ensure more efficient and specific targeting of unlawful behaviour and

crime occurring at sea. These sea operations will increase the maritime law enforcement capacity of those countries by EUR 425,600 and allow for more cost efficient and targeted responses to such activities at sea.

The **United Nations Office on Drugs and Crime** announced FishNET, a four-year EUR 4 million programme launched in January 2017 to address associated crimes in the fisheries value chain. This will be achieved through legislative reviews, the development of resource guides and capacity building activities to strengthen the developing countries' criminal justice response and law enforcement capacity.

Airbus announced that it is designing, building and will operate an entirely funded new constellation of four Very High-Resolution optical satellites to be launched from mid-2020. This new constellation will enable the growth of its maritime surveillance capacities worldwide in order to become the Primary Commercial Maritime Domain Awareness services operator and system provider. Combining its proprietary-data with all other available maritime sources such as AIS, VMS or MetOcean models, Airbus is building on its expertise on big data, machine learning and predictive analysis, in order to support the maritime industry to increase knowledge, anticipate threats, trigger alerts and suspicious behaviour and improve efficiency at sea. Airbus is therefore supporting governments to protect their Exclusive Economic Zone and private companies to secure their fleets against potential threats or source of illegal activities such as illegal fishing, smuggling or piracy.

Italy announced that it will support, until December 2018, Operation Atalanta, a EU counterpiracy operation focused on the Horn of Africa, providing the flagship and about 200 military personnel, for an overall economic effort of EUR 26.8 million. Italy announced the national maritime surveillance operation, Operazione Mare Sicuro, involving in the central Mediterranean Sea up to four Naval Units, airborne patrollers and 700 military personnel, for an overall economic effort of about EUR 84 million to be extended in 2018. The operation will carry out activities of protection and security of the maritime environment, ensuring freedom of navigation and protecting vessels involved in migrant rescue operations.

Italy announced that its Coast Guard will conduct within 2018 the EU-funded feasibility study for the full design of an Interagency National Coordination Centre under the control of the Ministry of Interior and a Maritime Rescue Coordination Centre under the control of the Ministry of Defense. Italy will also provide assistance to the Libyan authorities in providing the Libyan Maritime Rescue Coordination Centre with adequate Standard Operation Procedures.

Italy announced that its Coast Guard will conduct within 2018 the EU-funded feasibility study for the full design of an Interagency National Coordination Centre under the control of the Ministry of Interior and a Maritime Rescue Coordination Centre under the control of the Ministry of Defense. Italy will also provide assistance to the Libyan authorities in providing the Libyan Maritime Rescue Coordination Centre with adequate Standard Operation Procedures.

Italy announced that its Coast Guard will conduct within 2018 the EU-funded feasibility study for the full design of an Interagency National Coordination Centre under the control of

the Ministry of Interior and a Maritime Rescue Coordination Centre under the control of the Ministry of Defense. Italy will also provide assistance to the Libyan authorities in providing the Libyan Maritime Rescue Coordination Centre with adequate Standard Operation Procedures.

Italy announced a training program to be carried out within 2018 for up to 140 Libyan Ministry of Interior Coastal Police officers, and will provide up to six patrol boats to the Libyan Coastal Police. Italy announced a EUR 200 million fund for the implementation in 2017 of extraordinary short-term measures to increase bilateral cooperation with a number of African countries involved in illegal migration flows, at sea as well as on land to address the root causes of cross-sea migrations, including assistance to Tunisian authorities on maritime border control training and maintenance of naval assets.

Italy announced that it will support, in the 2017-2018 timeframe, Operation Sea Guardian, a NATO Maritime security operation focused on the Mediterranean Sea, with one Naval Unit, one maritime patrol helicopter and a submarine, for an overall economic effort of EUR 17.5 million.

Italy announced that it will support, until December 2018, Operation Atalanta, a EU counterpiracy operation focused on the Horn of Africa, providing the flagship and about 200 military personnel, for an overall economic effort of EUR 26.8 million.

Japan announced its commitment to support the space-based observation networks by the World Meteorological Organization (WMO) program by launching the geostationary meteorological satellites and adopted the budget of EUR 28,000 (USD 32,900) for Fiscal Year 2017 to operate Japan Meteorological Agency's Geostationary Meteorological Satellites "Himawari-8" and "Himawari-9". These satellites will provide, for a long term, stable meteorological observation information to more than 2 billion people to ensure their security and safety and lead to the improvement of meteorological services in the fields such as climate monitoring, natural disaster prevention/risk reduction, safe maritime transportation in the East Asia and Western Pacific region until 2029.

The **Japan Coast Guard** announced allocation of approximately EUR 633,000 (USD 700,000) for the period between April 2017 and March 2018 for the cooperation with and the assistance to its counterparts in Asia. One important example of this cooperation is "Coast Guard Global Summit" that took place on September 14, 2017. The Heads of Coastguards from 38 countries, region, and international organizations, from Asia, Oceania, the Americas, Europe, Africa and other regions in the world gathered in Tokyo. This is the first Summit of this kind.

Japan announced its commitment to fight piracy and armed robbery against ships through cooperation with UNODC (United Nations Office on Drugs and Crime). In 2017 Japan contributed almost EUR 20.8 million (USD 23 million) to UNODC, a 42.5% increase compared to 2016(footnote: This contribution is not limited to ocean-related activities).

Japan announced allocation of EUR 11.8 million (USD 13 million) in 2017 to fund the activities to enhance awareness of Tsunami in collaboration with the international

organizations such as UNISDR, UNDP and UNITAR as well as with JICA. The activities from April 2017 to March 2018 are related to "World Tsunami Awareness Day," established by the UN General Assembly resolution. Japan reaffirmed its commitment to disaster risk reduction (DRR) cooperation totaling EUR 3.6 billion (USD 4 billion) and to training of 40 000 personnel by the end of 2018, which includes projects for strengthening disaster risk management, enhancing capacity on weather observation and monitoring for earthquake, tsunami and flood, and earthquake early warning.

Japan announced allocation of approximately EUR 3.6 million (USD 4 million) for the period between March 2017 and February 2022 to the Technical Cooperation Project for Comprehensive Assessment and Conservation of Blue Carbon Ecosystems and Their Services in the Coral Triangle (Blue CARES), which consists in carrying out a comprehensive evaluation of Blue Carbon policies in Indonesia and the Philippines with the aim to propose a blue carbon conservation strategy in order to mitigate biodiversity loss and realize environmental improvements. This joint academic research project (Japan, Indonesia and the Philippines) will include the deployment of experts on Coastal and Marine Resources Management in February 2018.

Japan reaffirmed EUR 13.6 million (USD 15 million) support to the Japanese Technical Cooperation Project of Regional Initiative on Solid Waste Management (J-PRISM) in Pacific Island Countries Phase II (February 2017 to February 2022), which is based on the outcomes of J-PRISM Phase I (February 2011 to February 2016). This Project aims to strengthen both human and institutional capacity.

Japan reaffirmed its commitment to promote sustainable fisheries and announced its accession to the FAO Port State Measures Agreement in May 2017. Japan also announced its commitment to assist the relevant Regional Fisheries Management Organizations (RFMOs) through the acceptance of the agreement between Japan and the North Pacific Fisheries Commission (NPFC) regarding privileges and immunities of NPFC in June 2017. This Agreement will facilitate the work of the secretariat of NPFC, located in Japan.

Japan announced allocation of EUR 81.4 million (USD 90 million) concessional loan to Indonesia to rehabilitate its coastal area. This area is seriously affected by coastal erosion, seawall collapse and coastline recession.

Stop Illegal Fishing announced, on behalf of the **FISH-i Africa Task Force** to launch a programme of VIGILANCE in the Western Indian Ocean. This will be a robust assessment of all industrial fishing vessels licensed to fish by FISH-i members to assess the legality of their identity, history, ownership and management. This is in response to the systematic illegalities and criminal offences identified by FISH-i Africa investigations. This commitment will cover around 500 fishing vessels and will be completed by 2022, and is supported by the FISH-i Africa operational Task Force drawn from the eight member countries of **Comoros, Kenya, Madagascar, Mauritius, Mozambique, Seychelles, Somalia** and the **United Republic of Tanzania**.

The **Indian Ocean Commission (IOC)** announced the extension from 5 to 8 Participating States of the EU-funded Regional Fisheries Monitoring mechanism in the Southwest Indian

Ocean. The 8 Participating States (**Comoros, Madagascar, France / La Réunion, Kenya, Mauritius, Mozambique, Seychelles, Tanzania**) have signed the 2017 Fisheries Ministerial Declaration to commit to the institutionalised structure of the IOC Secretariat to combat IUU fishing and fisheries crimes in the industrial tuna fisheries in the South West Indian Ocean basin. This will promote a sustainable resource management of the second largest tuna fisheries producer worldwide.

Iridium Satellite announced its commitment to operate a Global Maritime Distress and Safety System (GMDSS) from 2020. This two-way safety communications system will aid in the coordination of urgent response to ships in distress. It also enables provision of information to ships about the safety of navigation, enabling cleaner and more efficient voyages.

Iridium Satellite announced its commitment to help to protect the marine environment by providing Long-Range Identification and Tracking (LRIT) services that can be used for identifying cases of pollution and unlicensed fishing, machine-to-machine (M2M) communications for real-time monitoring of oceanographic data, and related satellite-based services used in tracking marine hazards. In addition, Fisheries Vessel Monitoring Services (VMS) and Electronic Reporting Systems (ERS) will support catch reporting, which plays an important role in combating illegal, unreported, and unregulated fishing.

Kongsberg Satellite Services (KSAT) announced the establishment of a facility of up to EUR 830,000 (USD 1 million) to provide satellite services related to vessel detection. This will enable KSAT to assist agencies working on IUU fishing and threats to maritime security around the world. By combining data from a variety of satellite sensors and delivering the extracted information in near real time, KSAT is able to provide accurate and timely overview reports showing vessel activity across a large area of ocean, particularly highlighting those vessels which may be seeking to evade detection. Interested parties shall take the initiative to request this service.

Pew announced EUR 5 million (USD 6 million) additional funding, on top of its 2013 commitment of EUR 20.3 million (USD 24 million), to be fully committed by 2020 towards a global system of enforcement to fight illegal, unreported and unregulated (IUU) fishing. The additional funds will be invested in the adoption and implementation of policies, outreach to seafood markets, application of technology and law enforcement relating to IUU fishing. This will include continued work with partners such as FISH-i Africa on regional coordination, the FAO on implementing policies such as the Port State Measures Agreement, and OceanMind on the use of technology to detect illegal fishing activities and to provide a valuable verification tool for supply chains. We will also cooperate with national security agencies to help train the relevant maritime authorities.

Saildrone announced that by 2022 it will build a global ocean monitoring network consisting of 1,000 wind and solar powered saildrones providing planetary coverage at a 6x6 degree resolution, endeavouring to significantly reduce the cost of ocean data collection to improve management of sustainable fisheries, carbon cycle, marine pollution and climate change impacts on humanity. To further encourage scientific innovation, every year starting in 2018, Saildrone will award 30 days of saildrone-powered data, equivalent to approximately EUR

842,000 (USD 1,000,000) of research ship time, to the most impactful research project submitted.

Satlink announced that by 2020 its electronic monitoring software will be extensively enhanced for better fisheries control with image processing and the development of more filters in its video analysis for land-based observers. This will facilitate footage review, including potential discarding situations. It will further develop species' recognition in its satellite echo-sounder buoys to avoid bycatch.

Thales Alenia Space and its partners with the support of France announced that they will be investing EUR 26 million over 4 years to support the development of Stratobus - an autonomous stratospheric airship. The first flight of Stratobus is foreseen for early 2021. Stratobus will perform a wide range of missions ranging from (1) the detection of illegal activities such as illegal fishing and illegal ballast water discharge to (2) environmental monitoring such as marine pollution and the detection of harmful algae blooms and (3) maritime security such as search and rescue detection, anti-piracy, smuggling and migration. Various techniques and systems will be used: hyperspectral sensors and a radar/optical/AIS/VDES mission package.

Indonesia announced the Regional Cooperation Agreement's initiative to address crimes linked to the fisheries sector in the South East Asian Region such as money laundering, tax crime, corruption, human rights abuses, arms trafficking and illicit drugs trafficking.

Norway announced an initiative of NOK 39.6 million - approximately EUR 4.3 million - to combat transnational organised fisheries crime in cooperation with United Nations Office on Drugs and Crime. As part of this effort, Norway also supports the Interpol Fisheries Crime Working group. The project will run from 2017 to 2020.



Sustainable blue economy

The **European Union** announced that it will provide more than EUR 250 million to fund marine and maritime research in 2017. This includes EUR 40 million to support low-emission and advanced waterborne transport and over EUR 30 million for marine energy. Furthermore, the EU announced that it will provide EUR 12 million to support two new innovation projects on cleaning actions to combat marine litter and other pollutants. Finally, the European Union announced to support the BlueMED Initiative for cooperation on a healthy, productive and resilient Mediterranean Sea through science and research with over EUR 50 million.

The **European Union** announced that it is to further strengthen its work on the All-Atlantic Ocean Research Alliance by fostering enhanced cooperation frameworks with Atlantic partners such as Brazil and South Africa on marine science, research and innovation under the Belém Statement, and will allocate over EUR 60 million in the period 2018-2019 to fulfilling this objective. The EU will also continue to implement the groundbreaking Galway Statement on Ocean Research Cooperation with the USA and Canada. The European Union reported that the number of research teams working in international consortia on the challenges facing the Atlantic Ocean will exceed 500 by 2019.

The **European Union** announced that it will launch a EUR 14.5 million investment initiative in 2017 to promote a sustainable blue economy in the European Union. Around EUR 8 million of the fund is to provide start-up grants for high-potential projects in emerging blue economy sectors across the EU. In order to better monitor and combat marine litter, a further EUR 2 million will go towards providing support for innovative technologies to monitor and/or combat marine litter in waters around the European Union. Furthermore, EUR 3 million will go towards facilitating twinning projects in the Mediterranean Sea Basin, such as between maritime training and education institutes, businesses operating in the blue economy and local fishing communities. Finally, EUR 1.5 million is to be allocated to restoring marine and coastal ecosystems in the Mediterranean.

The **European Union** announced its intention to develop the Pilot Blue Science Cloud, which is to modernise the process of accessing, managing and using marine data, with the goal of improving the handling of large quantities of different marine and maritime data using cloud technologies. Furthermore, the Blue Cloud is intended to further foster work between EU scientists and their international partners. Cloud technologies can improve global and regional ocean observations and forecasting, as promoted in the framework of the G7 Future of the Sea and Oceans initiative and as part of the worldwide effort to build an improved Global Earth Observation System of Systems (GEOSS).

The **European Union** announced that it will finance the testing of the first wave and tidal array deployments in Europe in 2017 by contributing EUR 1.5 million to support administrations and project developers involved in environmental monitoring.

The **European Union** announced that it will provide EUR nearly 6 million to support projects in EU countries to set up cross-border cooperation on maritime spatial planning. Maritime

spatial planning works across borders and sectors to ensure human activities at sea take place in an efficient, safe and sustainable way.

The **European Union** announced that it will contribute at least EUR 1 million to support the World Bank's Global Fisheries Programme (PROFISH). The aim of the programme is to improve environmental sustainability, human wellbeing and economic performance in the world's fisheries and aquaculture, with a focus on the welfare of the poor in fisheries and fish farming communities in the developing world.

The **European Union** commits EUR 8.5 million to the preservation of marine and coastal biodiversity in the Caribbean Sea Basin for the benefit of communities that depend on these ecosystems. This action targets in particular natural areas that are threatened by misuse, overexploitation, pollution and climate change effects.

The **European Union** announced that it commits to further progressing Sustainable Fisheries Partnership Agreements with coastal states. These agreements already assist countries in the development of sustainable fisheries, the effective management of monitoring and control systems and the fight against IUU fishing. The new generation of agreements are to have a more integrated approach, including promoting a sustainable blue economy as well as advancing investment in the fisheries sector. This new approach should allow partner countries to gain more value from the ocean economy in a sustainable manner.

The **European Union** announced that it will invest EUR 23 million in the marine environment monitoring service of its satellite monitoring programme (Copernicus) in 2017 and 2018. The service focuses on climate change, fisheries and marine protection. It was also announced that Copernicus will, for the first time, create Ocean Monitoring Indicators, including on biochemistry. These indicators, important for measuring ocean health, will be published in the Ocean State Report that will be available online by the end of 2018.

The **European Union** announced the launch of the Pacific - European Union Marine Partnership (PEUMP) programme, worth EUR 45 million. Sweden announced that it will contribute EUR 10 million to the programme. The purpose of the programme is to support sustainable management and development of fisheries for food security and economic growth, while addressing climate change resilience and conservation of marine biodiversity.

On 24 March 2017, the **European Union** committed, in cooperation with the Intergovernmental Oceanographic Commission of UNESCO (IOC-UNESCO), to work on accelerating Maritime/Marine Spatial Planning processes worldwide. Maritime spatial planning (MSP) works across borders and sectors to ensure human activities at sea take place in an efficient, safe and sustainable way. Building on the Joint Roadmap, the EU will provide a grant of EUR 1.4 million to IOC-UNESCO to develop international guidelines for MSP. As part of this venture, two MSP pilot projects will be launched in early 2018: one in the Mediterranean and another in the South Pacific. Furthermore, an International Forum for MSP will be created to facilitate discussions on how MSP, including cross-sectoral actions, should be applied globally. The first workshop is to take place in spring 2018.

The **World Bank** announced a total of EUR 296 million (USD 352.5 million) over the next six years to advance the sustainable oceans and Blue Economy agenda in developing countries. This includes EUR 102 million (USD 121.4 million) to improve the management of fisheries in the South West Indian Ocean region (Madagascar, Seychelles, Maldives and the Indian Ocean Commission), with co-financing from the Global Environment Facility and Japan; EUR 34 million (USD 40 million) to innovate fisheries and aquaculture value chains in Peru; EUR 126 million (USD 150 million) for Vietnam to enhance coastal resilience; EUR 20 million (USD 23.6 million) for the Marshall Islands to strengthen early warning systems, support climate resilient investments in shoreline protection, and assist with crisis and emergency responses; EUR 15 million (USD 17.5 million) to improve solid waste management in the Maldives; and technical assistance (with financial support from Denmark) to help Indonesia develop an action plan to reduce marine plastic pollution.

The **European Investment Bank (EIB)** announced a EUR 92 million (USD 110 million) agreement with the Caribbean Development Bank to support many climate projects across the Caribbean. This loan follows a USD 65 million deal signed in 2011 to help nine projects in seven Caribbean countries. Over the past decade, the EIB has been helping the Caribbean fight problems related to climate change while also improving coastal ecosystems and reducing pollution. This assistance makes economic growth more sustainable in the region and supports the blue economy. These investments are known as Climate Action Framework Loans, meaning they support a wide range of projects. The loans not only help climate change mitigation, but they also support adaptation and resilience projects in energy efficiency, water infrastructure and physical and social infrastructures at community levels.

The **European Investment Bank (EIB)** announced to invest up to EUR 16.7 million (USD 20 million) in the Sustainable Ocean Fund to help fisheries, aquaculture, the seafood supply chain and coastal development, mainly in Latin America, Africa and Asia. The EIB is using its Climate Action and Environmental Facility for this financing. This facility is dedicated to projects that address climate change mitigation and adaptation as well as biodiversity and environment protection.

The **European Investment Bank (EIB)** announced a EUR 30 million loan agreement to the Amadéite Group, a French mid-cap company Amadéite - a pioneer in the field of marine biotechnology. The company develops algae-based health and nutrition solutions for animals, plants and humans, which help to limit the proliferation of algae and reduce the use of synthetic pesticides, fertilisers and antibiotics. Growth in algae hurts coastal environments as well as tourism. The loan is backed by the European Fund for Strategic Investments. This fund was started by the EIB and the European Commission to increase investment in areas such as digital technology, transport, renewable energy and innovation.

Royal Caribbean Cruises Ltd (RCL) and **World Wildlife Fund (WWF)** announced a new partnership on ocean conservation, grounded in ambitious and measurable sustainability targets. As a result of this partnership, RCL will by 2020 reduce by 35 percent its greenhouse gas emissions from a 2005 baseline; increase the procurement of sustainable seafood certified by the Marine Stewardship Council (to 65% by 2020) and Aquaculture Stewardship Council (to 75% by 2020); offer education to its 60,000 crew members and 5,000,000 guests on ocean conservation; and offer 1,000 tours certified to the Global Sustainable Tourism Council

standard to consumers visiting RCL's 500 worldwide ports of call. This partnership will also support the development and implementation of sustainable tourism guidelines for shark and ray tour operators as well as community-based whale shark ecotourism in Donsol, a Philippines coastal village.

Naval Energies announced that it will also continue to develop sustainable ocean energy solutions for Islands energy security, investing in ocean thermal energy conversion (OTEC) to produce non intermittent marine renewable energy for tropical regions and offshore floating wind for extreme sea conditions. Naval Energies is committed to carry out OTEC projects in French overseas territories (Martinique, la Réunion, Tahiti, Mayotte), Caribbean Islands and South East Asia.

Naval Energies announced that it is committed to deploy up to 35 tidal turbines by 2020, in Canada, Japan, France, Ireland and the United Kingdom, representing 70MW. These first arrays will be a clear milestone in the development of the tidal energy technology and launch the commercial roll-out of the marine renewable energies sector. That will contribute to reach EU targets for renewable energy production.

Naval Energies announced that the world's first tidal turbine plant in Cherbourg will be operational in 2018 and will create 40 direct jobs, embodying the start of the industrial phase of ocean energies. With a maximum production capacity of 25 turbines per year, the future plant will ensure the assembly and maintenance of pre-commercial and future commercial arrays in France. It will also operate as an industrial base for other pre-commercial projects around the world.

The **Nature Conservancy** announced that it will continue its work on innovative finance for the oceans. Following its successful model in the Seychelles, the Nature Conservancy is developing debt conversion programs including in the Caribbean, with an expectation to close new transactions in 2018. Financed by a Blue Bond with impact capital raised from private investors, and supported by credit guarantees, these efforts would target more than 200,000 square kilometres of new MPAs and upwards of USD 200 million in funding for their management and adaptation to climate change. Furthermore, The Nature Conservancy will actively explore new opportunities in regions such as the Indian and Pacific Oceans to leverage these financing solutions at a global scale.

The **Nature Conservancy** announced an investment of EUR 16.8 million (20 million USD) until 2022 to map the value of the oceans. Measuring value is crucial for stakeholders to understand how much they depend on ocean health, and thus to encourage appropriate management strategies.

The **Nature Conservancy** announced the development of an insurance product to protect a 30 KM long coral reef along the Mexican coast. The reef plays a key role in ensuring sustainability in the Mexican tourism sector. According to the Nature Conservancy, at least 102 nations and territories depend on coral reefs for income e.g. tourism and/or for protection e.g. from floods and tidal waves.

Malta announced the establishment of a new government body, with the remit of studying, protecting, preserving, managing and educating on the country's rich underwater cultural heritage with a budget of EUR 1.6 million over 3 years.

Malta announced the launch of a the Small States Centre of Excellence that will provide practical support to government administrations of small states to attain the Sustainable Development Goals under Agenda 2030. The Centre will provide relevant knowledge based tools, practical solutions and share best practices to leverage a collective synergy, catalysing the development of small states in the context of their individual limited resources. This will be done primarily through the use of the Internet and online based services. The website will be designed around the universally-accepted UN SDG Knowledge Platforms, particularly SDG goal 14.1 and 14.2 (marine pollution and marine protected areas).

Malta announced to designate 30% of its waters as Marine Protected Areas in 2018 to ensure protection of caves and reefs. As part of this objective, Malta announced it would develop management plans by 2020 for the fourteen marine protected area sites it designated in 2016 comprising roughly 3487 square kilometres, covering an area significantly larger than the country itself.

Malta announced it will commit to introduce a beverage container refund scheme by the end of 2019 to ensure that at least 70% of the plastic bottles generated in its islands are recovered and mitigate the impact of marine litter on the ocean.

Malta announced it would commemorate the pioneering role of the country in the creation of the UN Convention on the International Law of the Sea. The process within the United Nations General Assembly started in 1967 with the well-known speech of the Maltese Ambassador Dr Arvid Pardo. Celebrations will include a national symposium dedicated to the initiative by Dr Pardo in 2017, a side event at the UN on ocean governance in 2018 and annual lectures and promotional material (including dissemination of clips and documentaries shown on TV on a yearly basis) to underscore the relevance of his work and competitions on the Law of the Sea aimed at children.

Australia announced the building of a tool for the implementation of government policies by 2021 to guide and prioritise the Australian Government's activities to protect the environment. The Australian Government is committed to making ongoing improvements that enhance the currency and usability of this resource and will be exploring how it can make data available more frequently. Australia announced the release of the State of the Environment report as an interactive digital platform which enables decision-makers, researchers and the public to explore, compare, filter and download its findings in a variety of ways. This tool is an important resource for helping to guide and prioritise the Australian Government's activities to protect the environment, including the Marine environment. The Australian Government is committed to making ongoing improvements that enhance the usability of this resource and will be exploring how it can harness the value of making data available more frequently.

Australia announced that its national research agency, CSIRO, is leading a project, with a budget of EUR 1.33 million (AUD 2 million), from 2017 -2020. Its objective is to use field sampling and mathematical modelling to document the distribution of plastic in the ocean on

the coast and in the nearshore environment generated by 6-8 major urban centers and surrounding areas that have been identified as having significant waste mis-management or losses into the marine environment.

Australia announced the update of the threat abatement plan prepared in 2009, by mid-2018 aiming at providing national guidance on specific action to prevent and mitigate the impacts of marine debris. Injury and fatality to vertebrate marine life caused by harmful marine debris was listed as a key threatening process under Australia's Environment law. The plan update addresses six objectives, including the removal of existing marine debris and the increase of public awareness of the issue.

Australia announced a series of initiatives within the International Partnership for Blue Carbon, including a workshop in the Philippines in August 2017 helping the Coral Triangle countries (Philippines, Malaysia, Indonesia, Solomon Islands, Timor Leste, PNG) identify needs and opportunities to develop tools supporting knowledge sharing. In addition, Australia is hosting visiting fellows from IORA countries (Madagascar, Mauritius and Seychelles) in November 2017 and supporting a proposed IORA summit in 2018.

Australia announced EUR 10 million for the Integrated Marine Observing System (IMOS), launched in 2006 as a national collaborative research infrastructure. IMOS routinely operates a wide range of observing equipment throughout Australia's coastal and open oceans, making all of its data accessible to the users.

Australia announced EUR 24.7 million (USD 29 million) to the Climate and Oceans Support Program in the Pacific (COSPPac), supporting Pacific Island countries to adapt and mitigate the impacts of climate variability. COSPPac is successfully building capacity in Pacific Island countries National Meteorological and Hydrological Services and relevant Lands and Survey Departments to develop and disseminate user-focused products and services that assist governments and communities to better prepare for severe climate events. The program has ensured that the Pacific regional partners are equipped for the transition and end of the current phase of COSPPac in June 2018.

Australia announced EUR 204,000 (USD 240,000) to create a bilingual web-based tool to improve fish identification and fisheries monitoring. Available in 2020, the tool will have a wider South-East Asian application. All fisheries monitoring programs require accurate fish identification and fisheries assessments routinely rely on fish catch data, as recorded by port-based enumerators and/or on-board observers.

Australia announced EUR 300,000 (AUD 447,000) for a series of events to foster greater blue carbon awareness and cooperation in the Indian Ocean region. As a first step, Australia will hold a symposium in 2018 to develop a common understanding of blue carbon and a framework for future priorities in the Indian Ocean region.

Australia announced EUR 2 million (AUD 3 million) for the Blue Economy Challenge and "Aquacelerator" accelerator program to support the aquaculture industry, develop local economies across the Indian Ocean region and connect inspiring innovators with the networks capable of turning their ideas into reality.

Australia announced EUR 2.8 million (USD 3.29 million) for a four-year investment to increase capacity in Pacific island countries (PICs) to prevent, deter and eliminate illegal, unreported and unregulated (IUU) fishing in the region. The Niue Treaty Subsidiary Agreement (NTSA) is a multilateral treaty to strengthen fisheries management and provide for more cost-effective and efficient maritime surveillance in the Pacific region. Currently the NTSA has ten parties and entered into force on 30 July 2014. This project will support PICs through the early technically-demanding stages of ratification and implementation of the NTSA by: (i) enhancing regional information sharing and cooperation; (ii) supporting risk-responsive tasking of assets; and (iii) increasing capacity to undertake enforcement operations.

Australia announced EUR 1.3 million (USD 1.6 million) for the Pacific Maritime Boundaries Project which aims to assist Pacific island countries (PICs) to secure maritime jurisdictional rights consistent with international law. Running from 2017 to 2020, the project will focus on two main areas: technical and legal support to the Pacific Community (SPC) and PICs to finalise outstanding Pacific maritime boundaries and limits; and establishing an administrative infrastructure and capability required for boundary dissemination and which also supports the efficient use of marine space and the resolution of conflicts between users.

Australia announced EUR 238,000 (USD 280,000) for "A New Song for Coastal Fisheries - Pathways to Change" (New Song), addressing coastal fisheries management, a new approach developed with Australia's DFAT funding in 2015. The main aim is to strengthen the regional community-based ecosystem approach to fisheries management (CEAFM) as outlined in the "New Song" strategy, while ensuring communities have a voice in this process at all levels. The project aims to empower and proactively engage communities through better representation and coordination of CEAFM assistance utilising mechanisms that facilitate a coordinated region-wide approach. The project will work closely with partners and NGOs to strengthen existing networks for gathering and sharing information on "who is doing what where", and then map this out in collaboration with the partners.

Belgium announced the revision of its national marine spatial plan for the Belgian part of the North Sea. This revision is undertaken in order for the plan to be as up-to-date as possible and to ensure the coordination of decisions having a spatial impact on the Belgian marine waters with all stakeholders.

Belgium announced that by the end of 2017 it will adopt its federal action plan to combat marine litter, the scope of which will be broad and will include macro and microliter, land-based and sea-based sources, litter from the fisheries sector, single use plastic materials and primary microplastics. It will foresee cleaning and awareness raising activities. It will also focus on the collaboration on the national and international levels.

France announced its commitment to double the current volume of blue economy activities to EUR 104 billion and bring the number of jobs up to 600,000 over the next ten years. To support this objective, France has developed a method, based on a public-private partnership involving public authorities and the private sector.

France announced that it will ratify the International Convention on Standards of Training, Certification and Watchkeeping for Fishing Vessel Personnel (STCW-F) by 2019. This is part of France's commitment to improve working conditions and training in the shipping and fisheries sectors

France reaffirmed its commitment to develop a national strategy combining the implementation of the EU marine framework directive and the EU maritime spatial planning directive, reflecting and reconciling both environmental and economic development strategies along each of their four maritime facades.

Ireland announced its commitment to invest EUR 1 million over the period 2018-2021 into R&D to help ensure future food and nutrition security and the sustainable supply for other high-added-value chains and products such as bioenergy, pharmaceuticals and cosmetics from algae.

Ireland announced EUR 1 million funding towards research vessel ship time per annum from 2018-2021 to activities agreed under the Atlantic Ocean Research Alliance (AORA) - Galway Statement implementation, notably seabed mapping & ocean observation. Data from these activities will be open access.

Ireland announced that it will invest EUR 4 million per annum to a world class programme of seabed mapping (INFOMAR). Ireland has to date committed some EUR 50 million to this programme. This data feeds directly into the European Marine Observation and Data Network (EMODNnet) and helps support AORA.

Ireland reaffirmed that by end 2018 it will introduce national legislation to prohibit the sale or manufacture of certain products containing microbeads including not just cosmetics but also body care and cleansing products as well as detergents and abrasive surface cleaning products.

Ireland announced that, in cooperation with the International Eco Schools programme, it will roll out a ground-breaking Global Citizenship: Marine Environment education module for school children from September 2017 onwards. The programme will increase ocean literacy by fostering understanding of the important role the oceans play in our lives, how individual actions can affect them and how we can act together to protect them.

Ireland announced EUR 320,000 in funding to the Clean Coasts Programme for 2017 and commits to funding on an on-going basis in future years. This programme, operated by An Taisce, Ireland's National Trust, engages communities to work actively with local authorities to protect Ireland's beaches, seas and marine life. This Programme comprises Clean Coasts Volunteering (with over 550 Clean Coasts volunteer groups established to date) and the highly sought after Green Coast Award (62 beaches awarded the Green Coast award in Ireland in 2017).

Ireland announced the expansion of its 2015 "Fishing for Litter" programme to a further 2 ports/piers and 15 vessels which has encouraged fishermen to retain marine litter brought up

in their nets for onshore disposal, with 46 trawlers in 7 different ports around the Irish coast now participating in that programme.

Ireland announced a contribution of EUR 50,000 to the three United Nations trust funds in 2017. The contribution will go to the United Nations trust fund created to assist developing states engage with the Commission on the Limits of the Continental Shelf, to the trust fund to assist them in participating in the work of the UN Regular Process for global reporting and assessment of the marine environment and to the fund to assist developing states participate in the work of the UN Preparatory Committee on a new legal instrument for the conservation and sustainable development of marine biological diversity in areas beyond national jurisdiction. Since 2003 Ireland has contributed more than EUR 800,000 to these trust funds.

The **United Kingdom** announced EUR 9.1 million (GBP 8 million) between 2017 and 2018 to support Small Island Developing States (SIDS) to preserve their marine environments and tap into maritime resources to catalyse economic development in a sustainable way across Commonwealth countries. This is an addition to a package of EUR 6.3 million (GBP 5.6 million) announced in November 2015.

The **United States** announced that it has provided a 50% loan portfolio guarantee to Althelia's Sustainable Ocean Fund through USAID's Development Credit Authority for up to EUR 42 million (USD 50 million) in private debt investments in sustainable fisheries, supply chain improvement, and marine conservation. The Fund plans to invest in 10-20 sustainable businesses and organizations throughout Latin America, Africa, and Asia. The capital provided will help businesses create operational and behavioral change for sustainable fisheries activities, which will generate long-term revenue streams for those involved in the fishing industry and help improve coastal ecosystems.

The **United States** announced a partnership with the Walton Family Foundation to jointly support long-term conservation work in the biologically-rich Bird's Head Seascape area of the West Papua and Papua provinces of Indonesia through an anticipated EUR 4.2 million (USD 5 million), three-year grant to the Blue Abadi Fund. This fund will link various donors with the government of Indonesia.

The forty-three Union for the Mediterranean countries announced the launching of the Virtual Knowledge Centre (VKC) on Blue Economy, as a regional networking platform on marine and maritime affairs, with a view to consolidating the Mediterranean Blue Economy community by the end of 2018.

The forty-three Union for the Mediterranean countries announced their agreement to set up by February 2018 a Task Force on Environment in order to facilitate the implementation, among others, of the H2020 Initiative for a Cleaner Mediterranean, which is joining efforts of all committed stakeholders in addressing the 80 per cent of the sources of pollution of the Mediterranean Sea by the year 2020. They agreed as well to define priorities, operational modalities and a work programme for depollution and pollution prevention of the Mediterranean Sea for the post-2020 period.

The **United Nations Department for Economic and Social Affairs (UN DESA)** and **International Seabed Authority (ISA)** reaffirmed that during the period 2017-2018 they will hold a series of capacity building meetings/summits/workshops on Blue Economy in the Pacific which will further explore and highlight the potential contribution of the deep-sea mining industry to support national and regional development priorities. This is part of the on-going partnership between ISA, UN DESA and Pacific Small Island Developing States (PSIDS) to promote capacity building initiative for PSIDS.

Althelia Ecosphere, Aviva Investors, the BPCE Group, the European Investment Bank, Seventure Partners, Willis Towers Watson and the **World Bank** announced their participation in a partnership to support the development of a sustainable blue economy. Contributing towards this goal, they agreed to develop a set of sustainability principles that will guide investment and financing decisions in the 'blue economy', with a view to announcing these principles in 2018. They are also willing to engage with other financial institutions and economic players to promote this partnership. This partnership is supported by the European Commission, the World Wide Fund for Nature and the Prince of Wales's International Sustainability Unit.

Calysta is an innovator in sustainable products for the aquaculture industry. FeedKind® protein is a safe, natural, sustainable and traceable ingredient that provides a healthy alternative to wild catch-derived fishmeal. Calysta reaffirmed that annual production of FeedKind® protein will reach 20,000 tons by 2019 and 200,000 tons by 2020 without negatively impacting agricultural or water resources on land. Increased use of this type of feed in aquaculture can reduce fishing pressure on wild fish, leaving them to support the oceans natural ecosystems.

Conservation International announced that they are developing a Blue Accelerator Fund, an investment vehicle designed to support enterprises operating within the marine fisheries, aquaculture, marine tourism, and other ocean-related industries. Impact finance has the potential to accelerate and scale conservation outcomes, but is limited by the 'investment readiness' of enterprises which often lack key technical capacities, operational maturity, or sufficient commercial track record. The Accelerator Fund will address this gap by providing critical bridge financing and linking enterprises with investors, technical expertise, and markets for sustainable products and services. The Accelerator Fund aims to unlock larger-scale private capital investment by working with partners from initiation to exit, and will support protection of natural capital stocks and sustainable livelihoods in these critical sectors.

The **Natural Capital Coalition** and **Conservation International** announced they are building a coalition of organisations to produce an Oceans Supplement to the Natural Capital Protocol - an established framework for businesses to evaluate their impacts and dependencies on nature - by 2018. The Supplement will help a diverse set of companies to assess how their businesses depend on healthy oceans, identifying cost savings, new business lines, and improved access to financing; anticipating regulatory changes; and aligning with the UN Sustainable Development Goals. This process will receive inputs from corporate, finance, civil society and government partners, as well as leaders from ocean-based initiatives. The Coalition, which evolved from the TEEB for Business Coalition, is a diverse partnership of almost 250 organizations including businesses such as Dow Chemicals, Skanska, and

Walmart. Total resources of EUR 800,000 (USD 960,000) are needed to successfully deliver this project. In-kind resources of EUR 250,000 (USD 300,000) have already been committed.

DNV GL announced, in line with its vision of "Global impact for a Safe and Sustainable future", to address global aquaculture challenges related to competition for space, fish health and welfare, feed efficiency and environmental concerns by funding a dedicated research program of EUR 1 million for sustainable aquaculture, in co-operation with the industry. Their research will focus on how a digital transformation can make aquaculture more sustainable by providing advanced insight into both technical and biological performance. DNV GL also announced to make the main results of this research program publicly available by the end of 2018.

Federazione Imprese and **CONFALIS**, representing more than 30,000 companies with 200 Italian branches and 300,000 operators, announced the launch of the Foundation CSMARE in July 2017, as a specifically purpose-designed tool of private origin and public partnership for the promotion and support of strategic actions aiming at reducing pollution and preserving maritime ecosystem. Federazione Imprese and CONFALIS allocated EUR 15 million by 2022 to the Foundation for activities related to alternative fuels for the maritime transport, namely LNG and bio-LNG, the latter in particular produced from urban wastes and waste products of fish supply chain and agriculture.

Grimaldi Group announced the programme to build a new generation of roll-on/roll-off vessels 'Grimaldi's fifth Generation of Green ships' (G5GG), as part of a wider total investment EUR 2 billion (USD 2.3 billion) on newbuildings, ship lengthening, and retrofits over a four-year period, in new buildings, destined to short sea traffic with a 100% increase in cargo capacity against best existing ships. This will introduce an unprecedented sum of green technology for this ship category, including electronic propulsion engines connected with abatement systems of exhaust gas emissions. These are the most innovative energy saving devices, including air lubrication systems, solar panels and large Li-ion battery pack to enhance zero emissions in port. In this way, Grimaldi Group aims at achieving by 2020 an emission reduction of 50% per cargo transported at sea on short-sea route between North Italy and Spain and later on continental national cabotage and on Scandinavian routes

The **Intergovernmental Oceanographic Commission of UNESCO (IOC)** reaffirmed its global programme to enhance global cooperation on ocean literacy. With the support of Sweden, the IOC-UNESCO programme, will dedicate EUR 211,975 (USD 250,000) by the end of 2018 to develop an ocean school programme to foster ocean literacy among youth ages 10-18, an online platform to share resources, projects and people on ocean literacy for sustainable development and a comprehensive ocean literacy training programme for business, policy, and education sectors. Moreover, an international forum on ocean literacy, with 100 international experts will be held in December 2017. The project will be based on two main pillars: (i) Encourage cooperation and exchange on ocean education to improve ocean literacy globally and (ii) Seek and apply innovative ways to make citizens ocean literate.

The **Intergovernmental Oceanographic Commission of UNESCO (IOC)** and its partners reaffirmed the launch of the International Decade of Ocean Science for Sustainable

development in January 2018. This ten-year cooperative programme will support effective ocean management, stewardship and sustainable development, by: (i) Stimulating a global partnership on the marine science requirements needed to support implementation of Agenda 2030; with regionally driven focuses; (ii) Improving the understanding of the impacts of cumulative stressors and seeking sustainable solutions for sustaining benefits from the ocean; (iii) Sharing knowledge and enhancing interdisciplinary marine research capacities particularly for Small Island Developing States (SIDS) and Least Developed Countries. IOC/UNESCO is dedicating EUR 635,927 (USD 750,000) in 2017-2018 to prepare an implementation plan in consultation with Member States, UN and non-UN partners, as well as with the scientific community and relevant stakeholders.

IOC-UNESCO reaffirmed its commitment to dedicate EUR 3,815,564 (USD 4,500,000) for implementing two key initiatives for strengthening Coastal and Marine Ecosystem-based management through knowledge and information tools by March 2021: (i) With the support of the GEF the project "Strengthening Global Governance of Large Marine Ecosystems and Their Coasts through Enhanced Sharing and Application of LME/ICM/MPA Knowledge and Information Tools" or LME:Learn at a global scale; (ii) With the support of the Government of Flanders, Kingdom of Belgium the project "Implementation of an integrated approach to coastal and marine ecosystems for sustainable blue growth in the Southeast Pacific" or SPINCAM3 at regional scale. The objective of these two initiatives is to develop a harmonized coastal and marine environmental management framework for the beneficiary countries and large marine ecosystems, thereby providing the right tools to apply the ecosystem approach in their national waters, while also providing a foundation for planning sustainable economic development (blue growth) through inter-institutional capacity-building, which will translate into concrete recommendations and strategies at the regional level.

The **North Adriatic Sea Port Authority (NASPA)** announced its commitment to promote the renewal and expansion of the Venice Blue Flag agreement in 2018 and 2019; to reduce particular matter cruise emissions of more than 40 per cent at a local scale by using MDO/MGO fuel - instead of HFO as from entering the Venice Lagoon, through the Lido Inlet - and not only during hoteling phase- as foreseen by law; to evaluate, using Tier 3 EMEP/EEA (EEA 2009 - March 2011) methodology, the cost benefit analysis of Venice Blue Flag implementation; to publish online all data related to Venice Blue Flag results. The "Venice Blue Flag" consists in a voluntary agreement that has been signed, year by year, between local authorities and cruise companies, in order to encourage the use of green fuel (with Sulfur content less than 0.1%) within the Venice Lagoon. In 2017, all cruise companies scaling at the port of Venice signed this voluntary agreement. NASPA will also promote the Blue Flag agreement among other ports in Italy and the Adriatic region.

SeaEurope, the European maritime technology industry sector, announced the launch of a 4 years campaign from June 2018 to deploy, demonstrate and promote on-board and on-shore ballast water treatment solutions. With EUR 500 million of investment, the campaign aims to promote the implementation of the IMO Ballast Water Convention earlier than in 2024, by showcasing industry's early movers and virtuous conduct. Although the campaign has a worldwide reach, in 2018-2019 it will focus on the EU sea basins. The development of the campaign will be monitored through Key Performance Indicators such as number of systems retrofitted and volumes of untreated ballast water discharged averted from existing ships.

The **Sustainable Oceans Alliance (SOA)** announced EUR 4.27 million (USD 5 million) for the launch of the SOA Ocean Solutions Accelerator, a global program created to help young entrepreneurs build and scale ocean-focused start-up companies. The program will give young entrepreneurs access to skill-building workshops, funding opportunities, introductions to field experts, partners, mentors, and investors, as well as a platform to showcase their ventures on a global scale.

Indonesia announced a commitment of EUR 85,000 to develop several Special Integrated Blue Economic Development Zones. These projects will promote sustainable development through different initiatives based on blue economy principles. This will be related to its EcoPort project. Indonesia also announced that it will develop 200 fisher's villages and 416 fisheries port.

Indonesia announced EUR 85 million in support of the establishment of an Archipelagic and Island States Forum to strengthen cooperation among Archipelagic and Island States on the issue of blue economy, and climate adaptation.



Sustainable fisheries

The **European Union** has committed EUR 15 million under the PESCAO programme for the improvement of regional fisheries governance in Western Africa with the aim of developing a regional fishing policy, putting in place a regional coordination against illegal unregulated and unreported (IUU) fishing and improving fish stock management at regional level.

The fight against illegal, unregulated and unreported (IUU) fishing is a priority for the **European Union**. This is why the EU announced that it will contribute a minimum of EUR 1 million in 2017 to the FAO global programme to support the implementation of the landmark Agreement on Port State Measures to Prevent, Deter and Eliminate Illegal, Unreported and Unregulated Fishing. The programme provides policy, legal and technical assistance and capacity-building to strengthen enforcement of the Agreement. Furthermore, the EU announced that it will host the international conference to assess and review the Port State Measures Agreement in 2020. Finally, the EU announced that it will contribute EUR 225,000 in 2017 to FAO for the development of a global record that is to register fishing vessels, refrigerated transport vessels and supply vessels worldwide.

As a follow-up to the Medfish4Ever Declaration, a 10-year pledge to save the Mediterranean's fish stocks and protect the region's ecological and economic wealth that was signed on 30 March 2017, the **European Union** announced that it will commit EUR 5.7 million in 2017 to support the work of the UN Food and Agriculture Organisation (FAO) and the General Fisheries Commission for the Mediterranean (GFCM) in improving the sustainability of fishing resources in the Mediterranean.

The **European Union** announced that new rules are expected to enter into force by the end of 2017 to better and more sustainably manage the external fishing fleet. The new rules will allow the European Union to better monitor and control its fleet and efficiently address the problems of reflagging and chartering, thus enhancing efforts to combat IUU fishing..

The **European Union** is committed to reaching a multilateral agreement on fisheries subsidies at the 11th WTO Ministerial Conference that is to take place in Buenos Aires in December 2017. With this objective, the EU put forward a revised proposal in July 2017 at the World Trade Organisation to prohibit certain forms of fisheries subsidies which contribute to overcapacity and overfishing, to eliminate subsidies that contribute to illegal, unreported and unregulated fishing and to refrain from introducing new subsidies of this kind. The proposal, aimed at implementing SDG 14.6, also contains provisions on enhanced transparency and guidelines on special and differential treatment for developing and least developed countries. Furthermore, the EU will do its utmost to further this agreement and to support it through the stages of negotiation and implementation.

WorldFish announced a commitment of EUR 49,008,801 (USD 57,800,000) through the "Sustaining Small Scale Fisheries" Flagship of the CGIAR Research Program on Fish Agri-food Systems (FISH), jointly with the International Water Management Institute, James Cook University, the Natural Resources Institute at the University of Greenwich, and Wageningen

University, to ensure by 2022: (i) 1 million households have reduced poverty as a result of adopting improved fisheries management; (ii) 2.1 million hectares of aquatic and coastal marine habitat restored and under more productive and equitable management. Furthermore, the Flagship will assist 1.2 million people, of which 50% are women to exit poverty through livelihood improvements.

Mauritius reaffirmed its commitment to elaborate by 2021 a Marine Spatial Plan for the maritime zones of Mauritius to sustainably and effectively manage the conflict of uses in its waters.

Mauritius announced it will launch in 2018 an assessment of all the fishing vessels (including purse seiners and long liners operating in the region) operating in the waters of the Indian Ocean Commission States in order to study their identity, history, ownership and activities. With a view to prevent, deter and eliminate IUU fishing. Project's completion is expected in 2021. The data will be assessed and a report will be prepared with recommendations on means to combat effectively IUU fishing in waters of the Member States of the Indian Ocean Commission.

Mauritius announced the setting up of 20 nurseries for the major rehabilitation of the corals, on the top of the five already set up. The work is expected to start in 2018 and end by 2022.

Mauritius announced it will launch in 2018 an ocean data management project to improve data accessibility and provide a temporal and spatial perspective for the sustainable development of the ocean and its resources. The project is expected to complete in 2022 and is part of a substantial effort to harmonise and standardise ocean data so as to allow dissemination to the largest community possible.

Mauritius announced to increase research on the possible link of contaminated seafood consumption with the rise in diseases such as cancer by studying the accumulation of microplastics in the marine food chain and its potential impact on human health in the period 2017 to 2022.

The Government of **Nicaragua** announced that it is implementing the Resource management plan for the Queen Conch (pink snail strombus gigas) as of 2018. Catch quotas on production for the industrial and artisanal sector will be established, as well as quotas for commercialisation of the finished product. The Government of Nicaragua also announced that a proposal of a Nicaraguan Caribbean lobster management plan is being developed.

Nicaragua announced that by 2018 it will have its fisheries for lobster *spinosus* *panulirus argus* MSC certified (via traps).

Portugal announced it will develop new regulations by 2020 addressing the introduction of marine non-indigenous species to ensure the effective implementation of the Ballast Water Convention.

Portugal announced it will finance EUR 3 million by 2020 to support the upgrade of National Information and Communication Technology Systems on Ocean Environment Management Protection and Economic Activities.

Portugal announced it will increase by 2020 funding of deep-sea scientific research, including on seabed mining, with the creation of the Atlantic Observatory (based on the Azores Islands, an investment of 15 million euros) and the participation on the JPI Oceans Initiative.

By 2020, **Portugal** reaffirmed its commitment to develop technological platforms and tools that reduce marine pollution and promote the circular economy of the sea.

Portugal announced the expansion of the project "Fisheries for a Sea Without Litter" by 2030 to all fishing ports in the Portuguese mainland.

Portugal reaffirmed the intention to work at the regional level, within the OSPAR Convention, for the reduction of marine litter in the Atlantic.

Portugal reaffirmed its commitment to develop Maritime Spatial Plans (MSP) by 2020, covering the entire area under Portuguese jurisdiction and using ecosystem-based approaches.

By 2018, **Portugal** reaffirmed its commitment to create an evaluation and monitoring system of the Blue Fund that is aligned with the SDG 14 targets.

Portugal announced the reduction of by-catches and discards from fisheries by 2023.

Portugal reaffirmed its commitment for the conservation of at least 14% of coastal and marine areas under national jurisdiction by 2020.

Portugal announced the allocation of EUR 2 Million by 2020 to promote and increase the public investment on projects focused on marine biodiversity conservation.

Portugal reaffirmed its commitment to raise international awareness regarding "Oceans and Human Health"(already implemented and achieved through the organization of the International Ministerial Meeting of the Oceans Meeting 2017, in Lisbon, 7-8 September).

By 2020, **Portugal** reaffirmed its commitment to promote and support the launch of a marine and maritime research and innovation network, through the establishment of a Port Tech Cluster in Lisbon and the Atlantic Observatory in the Azores, with a focus on both North and South Atlantic;(a total investment of 30 million euros)

FAO reaffirmed the commitment made at the UN Ocean conference to strengthening fisheries governance and States' capacities to prevent, deter and eliminate illegal, unreported and unregulated fishing by providing EUR 11,8 (USD 14 million) by 2023 in technical support to developing States Parties. This will be achieved through a combination of global and regional programmes, national projects and ad hoc country level assistance executed by FAO in

partnership with concerned stakeholders, funded through the FAO regular programme and voluntary contributions from States and other partners.

FAO announced to commit mobilizing an additional EUR 2.5 million (USD 3 million) through FAO regular programme and voluntary contributions from States and other partners by 2020 to upscale its work with member countries, small-scale fisheries actors and other partners to raise awareness, strengthen institutional capacities, empower small-scale fisheries organizations, generate and share knowledge, support policy reform and provide technical assistance to support the implementation of FAO's Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries in the Context of Food Security and Poverty Eradication (SFF Guidelines) at global, regional and national level.

FAO announced to invest more than EUR 422,000 (USD 500,000) through an inter-regional TCP called the Blue Hope Project that will initially involve 3 coastal countries along the Mediterranean (Algeria, Tunisia and Turkey) under its Blue Growth Initiative. The project will focus on building coastal community resilience through improved technical capacity among government agencies and communities to sustainably manage their coastal resources, especially inshore fisheries, thus improving their food security and their livelihood opportunities as well as their capacity to respond to drivers of change such as climate and migration.

FAO reaffirmed its commitment made at the UN Ocean conference, to spend an extra EUR 295,250 (USD 350,000) to ensure that trade contributes towards the achievement of the UN Sustainable Development Goals by reinforcing the multilateral trading system and ensuring that trade policies and strategies are coherent with other enabling national policies. Funding will be used to convene three separate regional SIDS workshops on fish trade in 2018 – Pacific, Caribbean and AIMS, to keep SIDS in the three regions informed on current processes and outcomes underway to reach a positive outcome on fisheries subsidies, on trade governance and regulatory framework issues necessary for effective fish trade.

The **General Fisheries Commission for the Mediterranean of the FAO (GFCM)** announced EUR 19,559 million (USD 23 million) by 2020, under its Mid-term strategy towards the sustainability of Mediterranean and Black Sea fisheries to help implement requirements of SDG 14 and the 2030 Agenda. These funds will help reverse the trend of overexploitation of Mediterranean's iconic marine species, strengthening scientific advice for management while supporting the livelihoods of coastal communities. In addition, by investing in technological advances that level the playing field (regional VMS and control system), FAO will help countries implement effective port State measures to curb IUU fishing, and support mitigation of unwanted fisheries-ecosystem interactions through bycatch and discard reduction programmes."

The **General Fisheries Commission for the Mediterranean of the FAO (GFCM)** jump-started and spearheaded a proposal to declare an International Day for the Fight against Illegal, Unreported and Unregulated Fishing. The objective of this day will be to raise awareness at all levels, including civil society and general public, on the threats posed by IUU to the conservation and sustainable use of fisheries resources and its negative economic impacts in particular on developing countries dependent on fisheries resources. The FAO

conference adopted this proposal at its 40th session in July 2017 and requested the UN General Assembly to consider declaring 5 June as the International Day for the Fight against Illegal, Unreported and Unregulated Fishing. Pending the adoption of the UN General Assembly, GFCM commits to support the celebration of the ID-IUU through raising-awareness activities and dissemination of materials for the general public and civil society on the threats associated to IUU fishing.

The **ILO** announced EUR 1.5 million (USD 1.8 million) in 2018 to implementing projects to combat unacceptable forms of work and human trafficking in the fishing and seafood industry, including the Global Action Programme against forced labour and trafficking of fishers at sea (GAPfish), and to mobilise resources for new ones.

Microsoft co-founder and philanthropist **Paul G. Allen** deepened his commitment to ocean conservation and announced a EUR 34 million (USD 40 million) investment to develop and deploy a near-real time illegal fishing intelligence and research program called SkyLight. Mr. Allen's company Vulcan Inc. is committing its resources across technology, artificial intelligence, aerospace and public policy to assist governments in the global fight against illegal fishing, which threatens the world's food supply, marine ecosystems, and security. Early adopters of SkyLight include Gabon and Palau.

Oceana announced a major insurance industry commitment contributing to the international fight against illegal, unreported, and unregulated (IUU) fishing. Global insurance industry leaders including Allianz AGCS, AXA, Hanseatic Underwriters, Generali, and The Shipowners' Club have signed a statement committing to not insure or facilitate the insuring of IUU fishing vessels. Signatories have also agreed to encourage the adoption of other measures to help eliminate IUU fishing. The Statement has been developed in collaboration with the Principles for Sustainable Insurance, a global sustainability initiative of the United Nations Environment Programme Finance Initiative. The Statement remains open for signatures and can be viewed online at <http://eu.oceana.org/en/assisting-ocean-stewardship-through-marine-insurance>

Korea announced EUR 85 million (USD 100 million) between 2017 and 2027 for the establishment of the World Fisheries University, which pilot programme was launched in September 2017. The objective is to build capacity mainly in the SIDS and developing countries towards the development of sustainable fisheries.

Korea reaffirmed its commitment to invest EUR 4.5 million (USD 5.3 million) between 2017 and 2021 to support the establishment and operation of a Marine Acidification Monitoring system -a network to monitor acidification in coastal areas of Small Islands Developing States in the Pacific-. Korea will offer training on the operation of the acidification monitoring system to officials from Palau and the Secretariat of the Pacific Regional Environmental Programme (SPREP).

Korea reaffirmed its commitment to invest EUR 28 million (USD 33 million) annually to manage the marine environment around the Korean Peninsula. This includes monitoring the inflow of pollutants into the sea and restoring coastal ecosystems. Korea will also conduct research on marine litter and collect marine waste from the seabed.

In seriously pursuing its commitment for sustainable fisheries, the **Philippines** announced a major overhaul of its national fisheries law, specifically adding strong provisions that promote science-based fisheries management and better enforcement. Under the newly enacted law, the country's major fishing grounds are to have science-based management plans by 2022. Moreover, in 2017 the government has appropriated around USD 12.3 million for the expansion of the Vessel Monitoring System to include the domestic large and medium- scale commercial fishing vessels. In compliance with the amended law and on government resources, the VMS will be completed in 5 phases. The Philippines hereby commits to further expand the VMS to 35% of its registered fishing vessels by 2020.

Spain announced its commitment to develop and boost the sustainability of the blue economy by (1) financing at least 20 projects per year on this issue, (2) engaging 10,000 volunteers in coastal areas for clean-ups, (3) creating a national citizen science marine program, (4) supporting 1,000 entrepreneurs and (5) fostering the creation of 100 new enterprises and business lines linked to the blue economy framework. Lastly, different awareness raising actions on the marine protected areas' value will be carried out to reach 7 million Spanish (15% of Spanish population) by 2024.

Spain announced an annual financial support of EUR 50,000 FAO Global record until 2022.

Spain announced its commitment to preserve the artisanal fisheries sector and its ecological benefits in marine reserves. In order to achieve that, Spain will allocate EUR 150,000 in 2018 to continue to study the effects of Climate Change in marine reserves which play a crucial role as a tool for the conservation of biodiversity and fisheries management.

Spain announced a creation of new Marine Protected Area "Cetacean migration corridor of the Mediterranean" of 46,000 square kilometres (almost 5% of Spanish waters) in the Mediterranean Sea between the coasts of Catalonia, Valencia and the Balearic Islands. This MPA will be recognized under the Barcelona Convention. With this new MPA 13% of Spain's marine waters will be under protection.

The **United States** announced a partnership with the Walton Family Foundation, the David and Lucile Packard Foundation, and the Gordon and Betty Moore Foundation to jointly launch a Seafood Alliance for Legality and Traceability (SALT) with USAID awarding a EUR 4.4 million (USD 5.3 million) five-year cooperative agreement. This multi-stakeholder alliance of government, business, and civil society will facilitate knowledge exchange, innovation, and action to promote legal, sustainable fisheries by enhancing the ability to trace seafood supply chains. The Fisheries Agency of Japan announced its intent to participate in the SALT design process, bringing its leadership and expertise in fisheries management to this process.

The **United States** announced EUR 4.2 million (USD 5 million) for a program to combat forced labor and human trafficking on fishing vessels in the Asia-Pacific region. The project will work in two countries in the region and focus on two key outcomes. First, it will improve enforcement to combat labor exploitation, including forced labor and human trafficking, on fishing vessels through strengthened government regulations and policies, improved coordination within government interagency task forces, and increased use of multi-

disciplinary inspections that mainstream inspection for labor exploitation. Second, it will combat forced labor on fishing vessels through improved understanding of acceptable conditions of work and indicators of labor exploitation, improved ability to report exploitative working conditions, and improved access to remedy of exploitative working conditions on fishing vessels.

The **United States** announced the fourth annual Fishackathon, a public-private partnership with HackerNest for a weekend coding contest to address sustainable fishery challenges. Hundreds of technologists and entrepreneurs around the world will spend February 10-11, 2018, competing to build practical tech solutions to endemic problems defined by fisheries experts. For more information, and to see the winning submissions after the event, visit Fishackathon.co.

The **African Confederation of Artisanal Fisheries Professional Organizations (CAOPA)** announced its commitment to develop, in a participatory, transparent and gender-sensitive manner, at least one national - or where appropriate, regional - Plan of Action per year, for the implementation of the FAO Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries in the Context of Food Security and Poverty Eradication. This commitment is undertaken in preparation of the International Year of Artisanal Fisheries and Aquaculture, to be held in 2022, with the first meeting to be held in November 2017 where the content of national plans of action will be discussed and agreed by CAOPA members. CAOPA will mobilise funds and expertise from its partners (including CFFA, Bread for the World, Swedish Society for Nature Conservation) to develop these National Actions plans. Per year, an amount of EUR 100,000 will be allocated towards that end, i.e. EUR 600,000 until 2022.

Bolton Alimentari and **WWF** announced a new partnership to supply one hundred per cent of Bolton Alimentari seafood from sustainable sources by 2024. This partnership will support well-managed fisheries, foster transparency along the seafood supply chain and increase consumer's awareness on the importance of sustainable seafood. By sharing best practices, the partnership will inspire other seafood companies to embark on the road of sustainability. The partnership entails a financial commitment made by Bolton Alimentari of EUR 1,600,000 from 2017-2020.

Carrefour announced that half of all fish that it sells (both wild and farmed) will come from sustainable sources by 2020. To fulfil this commitment, Carrefour is working with its suppliers to select fish from healthy stocks, promote more selective catching methods, develop sustainable aquaculture by reducing antibiotics and using non GMO feed, and fight illegal fishing.

Earth Twine, a collaborative technology company that works to create and bridge leading edge technologies into traditional food supply distribution, announced the launch of the Earth Twine Blockchain Solution. This system developed specifically for the seafood industry is dedicated to sustainable seafood-oceans. It validates origin and production information of seafood by integrating smart contracts.

EcoHub announced it will launch the mFish initiative in Malaysia, Vietnam, Philippines, Thailand, Myanmar, Bangladesh and Maldives in their local languages, with the intent of

capturing more than 100,000 users. mFish provides mobile services to small-scale fishermen in developing nations through the mFish application. mFish is provided to fishermen at \$0 cost of data through Facebook's Free Basics platform, and allows them to access market and weather information and more easily report catch documentation for the purpose of traceability and fisheries management.

The **General Fisheries Commission for the Mediterranean of the FAO (GFCM)** announced EUR 19.56 million (USD 23 million) by 2020, under its Mid-term strategy towards the sustainability of Mediterranean and Black Sea fisheries to help implement requirements of SDG 14 and the 2030 Agenda. These funds will help reverse the trend of overexploitation of Mediterranean's iconic marine species, strengthening scientific advice for management while supporting the livelihoods of coastal communities. In addition, by investing in technological advances that level the playing field (regional VMS and control system), FAO will help countries implement effective port State measures to curb IUU fishing, and support mitigation of unwanted fisheries-ecosystem interactions through bycatch and discard reduction programmes."

The **GEF** announced a total of EUR 19 million (USD 22.3 million) investment in fisheries. This included EUR 1.45 million (USD 1.7 million) investment in fisheries in the Caribbean to be implemented by FAO and EUR 5.36 million (USD 6.3 million) investment to assist Caribbean countries to transition toward a blue economy to be implemented by the World Bank. The GEF announced EUR 6.2 million (USD 7.3 million) to address fisheries in West Africa to be implemented by the World Bank. In addition the GEF announced a EUR 6 million (USD 7 million) investment for a UNDP led initiative in the Humboldt Current to further sustainable ocean governance, particularly fisheries, in Chile and Peru.

International Seafood Sustainability Foundation (ISSF) announced that at least 75% of the tuna industry will follow new ISSF measures for sustainability best practices. These measures facilitate continuous improvement toward sustainable tuna fishery resources. It also announced that it audits compliance against such measures. Two recently adopted measures help fight IUU fishing and support sustainable management of tuna fisheries resources: ISSF now requires the use of non-entangling FADs by participating vessels and requires participating companies to identify all species of tuna and the area of capture by means of product labelling or through a publicly available web-based traceability system. The first compliance audit reports for these new measures will be published in 2019.

The **Marine Stewardship Council** announced its commitment to engage 20% of global marine catch in its program by 2020, supporting productivity and resilience in globally important marine ecosystems. This will be achieved by expanding the number of key Large Marine Ecosystems in which more than 20% of the marine catch is MSC certified, and growing the number of key commercial species, and fisheries in the Global South, engaged in the MSC program.

Nofima, the Norwegian Institute of Food, Fisheries and Aquaculture Research, announced the investment of EUR 1.5 million in the period 2018-2020 to improve socially and economically sustainable seafood production. This will be achieved through the development of models to handle and mitigate conflicts on the overuse of common marine resource. As part of this

work, Nofima will develop systems for systematic evaluation of value creation potential, profitability, logistic challenges, climate impact and market opportunities for industries related to the use of Arctic krill (*Calanus finmarcticus*), mesopelagic fish and value of by-products

The **Organization of Associated Producers of Large Tuna Freezers (OPAGAC)** announced that it will apply the standard for responsible tuna fishing (AENOR standard) and apply social conditions on board fishing vessels in line with the 2007 Work in Fishing Convention of the International Labour Organisation. These standards are to apply on board all of its vessels before the end of 2017. OPAGAC also announced that it has allied with the World Wildlife Fund on the Global Tuna Fisheries Improvement Project in order to achieve certification under the Marine Stewardship Council for all OPAGAC catches by 2021. OPAGAC will work to join both initiatives, in order to achieve a fisheries certification that includes environmental and social sustainability of the tuna fisheries as a Global standard.

The **Global Partnership for Sharks and Rays** announced that it plans to award EUR 6.75 million (USD 8 million) over the next two years to support shark and ray conservation around the world. These grants will be used in selected countries to protect endemic sharks and rays that are threatened with extinction; promote sustainable trade in shark and ray products through the implementation of management measures for species listed on Appendices I or II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES); and sustainable fisheries management through regulations adopted by regional fisheries management organizations (RFMOs) such as the International Commission for the Conservation of Atlantic Tunas (ICCAT).

Rare commits to mobilize EUR 84,729,248 (USD 100 million) by 2021 and capitalize a EUR 15,251,265 (USD 18 million) Impact Fund by October 2017 in support of sustainable small-scale fisheries; ending overfishing, protecting critical marine habitat, strengthening access for small-scale fishers to marine resources and improving economic and social resilience. Rare commits to building the capacity of local communities and partners, supporting national policy and value chain improvements to reduce the threat from overfishing in more than 32 million hectares of coastal waters and engage more than 1 million fishers in sustainable fishing practices in at least 10 countries globally.

Thai Union, one of the world's largest seafood companies, announced an ambitious strategy to ensure 100% of all of its tuna brands (including John West, Genova, Chicken of the Sea, Mareblu, Petit Navire and SEAELECT) are to be sustainably sourced with a commitment of achieving a minimum of 75% by 2020. Sustainably sourced tuna are defined by Thai Union as either already certified according to the standards of the Marine Stewardship Council (MSC) or are involved in a Fishery Improvement Project (FIP) that is working towards achieving standards required for MSC certification. Thai Union will invest 90 million USD (EUR 77 million) in initiatives to achieve this objective and allied with a number of partners including WWF, the Marine Stewardship Council, Legambiente and the Global Sustainable Seafood Initiative. In agreement with Greenpeace, Thai Union also agreed to a comprehensive package of reforms to improve tuna traceability and reduce bycatch

WWF announced a EUR 5 Million programme funded by a private foundation and delivered in partnership with the General Fisheries Commission for the Mediterranean, and other NGOs to transform small scale fisheries in the Mediterranean. This five year programme will work with small scale fisher-people across the Mediterranean to increase capacity and develop co-management schemes leading to sustainable fisheries management and improved livelihoods. This will entail a regional level engagement with fishers, authorities, and other stakeholders, as well focused efforts in selected Small Scale Fisheries sites in Italy, Greece, Croatia, and Turkey. The project will incorporate WWF's ongoing work in another ten fisheries in Algeria, Albania, Spain, France and Tunisia. All together, these eight countries represent more than sixty per cent of the Small Scale Fisheries sector in the Mediterranean.

Palau announced that through public-private partnership, it will open the first regional center for maritime surveillance in 2018 to address Illegal Un-reported Unregulated (IUU) fishing.

Norway announced that it will allocate up to NOK 10 million - approximately EUR 1.1 million - to help developing countries implement the Port State Measures Agreement to Prevent, Deter and Eliminate Illegal, Unreported and Unregulated Fishing, in partnership with FAO. The expected deliverable is to increase the capacity in developing countries to implement and enforce the provisions of the agreement. The first Meeting of the Parties to the Agreement took place in Oslo in May 2017.

Norway announced its support to a project to be implemented by FAO to enhance awareness of governments and other stakeholders at global, regional and national level on the contribution of small-scale fisheries to food security and poverty reduction. The project also aims to improve technical and organizational capacities related to small-scale fisheries in developing countries. The support will consist of up to NOK 23 million - approximately EUR 2.5 million - and run for several years.

Norway announced that it will establish a Global Action Network on Sustainable Food from the Ocean for Food Security and Nutrition under the UN Decade of Action on Nutrition 2015-2025. The purpose is to work towards the achievement of the Sustainable Development Goals and follow up on the outcome of the Second International Conference on Nutrition (ICN2). A network of countries will be established in 2017-18.

Thailand announced its commitment to participate in the Global Record of Fishing Vessels, Refrigerated Transport Vessels and Supply Vessels (global record) by 2018 to support the global implementation of Port State Measures. Thailand will make information available on its fishing fleets and enter it into the Global Record Database. Thailand also announced that it has entered into a progressive and technical cooperation with the Seafood Task Force and Ocean Mind to support safe and sustainable seafood as well as to prevent the entry of fish and fishery products from IUU (Illegal, Unreported and Unregulated) fishing activities into the supply chain; Thailand using new tools for a strict enforcement of Port State Measures.

Thailand announced that it has initiated cooperation on improving fisheries management and combating IUU fishing with ASEAN Member States through the development of ASEAN Common Fisheries Policy. Through the ASEAN Sectoral Working Group on Fisheries mechanism, Thailand aims to finalize the elements to be included in the Policy within 2019.

The Policy will strengthen ASEAN collective efforts for sustainable and responsible fisheries and food security.

Thailand announced that it will ratify and implement related ILO Conventions including Convention no. 188 on Work in Fishing to ensure labour standard at sea and that migrant workers are not at risk of harm.



Climate change

The **European Union** announced that it will commit EUR 1.5 million to reducing black carbon emissions in the Arctic. The project is intended to reinforce international cooperation to protect the Arctic environment.

The **European Union** announced that it will contribute EUR 600,000 over the next two years to an integrated Arctic project focusing on the three priority areas of EU Arctic policy: Climate Change and Safeguarding the Arctic Environment; Sustainable Development in and around the Arctic; and International Cooperation on Arctic Issues.

The **European Union** announced a EUR 10 million project with the International Maritime Organisation (IMO) concerning climate change mitigation in the maritime shipping sector. The project aims to establish five Maritime Technology Cooperation Centres (MTCCs), one in each of the target regions - Africa, Asia, the Caribbean, Latin America and the Pacific - thereby forming a global network. The network's task is to enable developing countries in these regions to develop energy-efficiency measures in maritime transport.

Iceland reaffirmed its commitment to update of its climate mitigation strategy by the end of 2017, aimed at ensuring that the country can meet its commitments under the Paris Agreement up until 2030. Iceland will reduce emissions from transport and fisheries by using low-carbon fuels and electricity and other means; green taxes and incentives; reduced waste and better waste handling; carbon capture; and increased carbon uptake by restoring forests, vegetation and wetlands.

Iceland reaffirmed its commitment to launch a ground-breaking 13-year effort to map the remaining 88% of its Icelandic Economic Exclusive Zone. Two high-tech platforms will be used for the work that will be operating beyond 200 meters using an ultramodern multibeam echosounder technique. The estimated cost of the project is around EUR 26 million (31 million USD). All data that is made available from the project will be available for use free of charge for non-profit use.

Microsoft announced today a dedicated "AI for Earth EU Oceans Award" as part of the AI for Earth program launched in July 2017, which has committed USD 2 million in the form of grants that provide access to Artificial Intelligence and cloud tools, and resources to use them well, to organizations working on the issues of water, agriculture, biodiversity and climate change. The "AI for Earth EU Oceans Award" will allocate some of these funds exclusively to work on oceans with EU-based research institutions. The award will provide access to Azure compute resources and data science tools to qualifying applicants that are focusing on addressing ocean-related challenges. Submissions are due by December 15, 2017; grant decisions will be communicated 4-6 weeks following this date; the grants are for a one-year term.

Sociedade Francisco Manuel dos Santos, the holding company of an international retail and agro-foods group, announced EUR 41 million over the next eight years to support ocean conservation and ocean literacy, with a primary geographical focus in Portugal, the Atlantic

Ocean, Portuguese speaking countries and Small Island Developing States (SIDS). This initiative is part of a ten-year commitment for a total amount of EUR 81 million, including a new philanthropic foundation set up in 2015. The Oceano Azul Foundation is dedicated exclusively to ocean sustainability and funding for a 30-year concession of the Oceanário de Lisboa. Its mission and its yearly profits will be dedicated to ocean conservation and education. Ocean Azul has launched a mass market campaign with TV ads raising awareness on plastic litter polluting the oceans. Since 2016, the Foundation is sponsoring a center for recovery of wounded marine species in the coast of Portugal with an annual grant of EUR 100,000.

The **Oceano Azul Foundation** and the **Waitt Foundation** reaffirmed their commitment to invest at least EUR 1 million in the next three years to help create and implement a network of effectively managed marine protected areas in the Azores archipelago. The project, named Blue Azores, is a partnership between the two foundations, the Portuguese government, regional administrations and different stakeholders and is to include evaluating the economic value of ecosystem services, organising scientific workshops, media and science expeditions, literacy and awareness campaigns and sustainable fisheries actions. The Azores include some of the most important marine habitats in the Atlantic Ocean, including coastal island habitats, the deep sea, seamounts, hydrothermal vents and the open ocean. The islands attract marine megafauna such as whales, dolphins, sharks, mantas, billfishes and tunas, and a hotspot for marine biodiversity.

The **Oceano Azul Foundation announced** announced a partnership with the **Environmental Defense Fund Europe** to promote conservation initiatives in Portugal's small scale fisheries. The Foundation has committed EUR 500,000 for the next 5 years, with further public and private funding to be raised, in order to empower local fishing communities to cooperatively manage their fisheries alongside public authorities, academic and scientific institutions and NGOs. The ultimate objective of the partnership is to protect fish stocks while improving fishers' livelihoods.

Carnival Corporation & plc announced the order of seven new LNG cruise ships to be delivered between 2018 and 2022. These new vessels will be part of the first generation of cruise ships to be fully powered by LNG both in port and on the open sea

France announced in accordance with the Paris agreement objectives on climate change that it will ban the production of hydrocarbon on its territory by 2040's.

The **Ocean and Climate Platform, France** and **UNESCO-IOC** announced the launch of the Ocean and Climate Initiatives Alliance (OCIA) in February 2017. France (the French Agency for biodiversity) has provided EUR 300,000 to the Alliance. Its ambition is to promote the integration of climate-related ocean initiatives into the Global Climate Action Agenda, and to reinforce ocean resilience by working to meet the UN Sustainable Development Goal 14. The Alliance's objectives include carrying the voice of civil society into international dynamics; reducing the knowledge gap on ocean and climate and, sharing existing scientific information; and identifying the synergies between international initiatives on ocean and climate. Accordingly, the Alliance will release its first report on the progress of ocean and climate action during the Oceans Action Day at UNFCCC COP23.

France announced an impact study of EUR 250,000 to consider establishing a low pollution emission area (Nitrogen Emission Control Area / Sulphur Emission Control Area) by ships in the Mediterranean.

France reaffirmed its commitment of EUR 800,000 to support eight research projects to advance scientific understanding of ocean acidification, and its impacts on marine biodiversity (coral reefs, phytoplankton etc).

France announced a EUR 1.5 million commitment to protect and restore marine and coastal ecosystems in the Northern Mozambique Channel. France also committed EUR 1.2 million to a marine restoration project in Costa Rica.

The **United Kingdom** announced EUR 11.4 million (GBP 10.1 million) between 2016 and 2023 to design and introduce a model of sustainable development for mangrove habitats. The project will focus on blue carbon sequestration and forestry management, fisheries management and improvement, mangrove livelihood diversification, community health and women's empowerment.

The **United Kingdom** announced EUR 5.88 million (GBP 5.2 million) for marine projects through the Darwin Initiative and Darwin Plus competitive grant schemes in 2017. Through these initiatives, the UK funds environment and biodiversity projects in developing countries and the UK Overseas Territories. These competitive grant schemes aim to build local capacity to manage biodiversity and the natural environment for the future, securing the benefits of these natural resources for people. A key priority for Darwin Plus is to help deliver a Blue Belt of marine protection around the Overseas Territories. This includes developing integrated marine management plans and promoting sustainable fisheries. Enhancing the extent and management of protected areas in the marine environment is also a priority for the Darwin Initiative.

The **United States** announced that the National Oceanic and Atmospheric Administration (NOAA) will expand the Pier2Peer scientific mentorship program from the current 49 active pairs of scientists to 60 pairs by the 2018 Our Ocean conference, with a special emphasis on the Pacific Island nations. NOAA launched Pier2Peer in April 2016 as part of the Global Ocean Acidification Observing Network (GOA-ON). The program pairs early-career researchers with experienced scientists, creating international networking and collaboration opportunities and increasing global scientific capacity.

The **United States** announced that, for 2017, it has allocated EUR 363,000 (USD 433,875) through the International Atomic Energy Agency's Peaceful Uses Initiative to the Ocean Acidification International Coordination Center (OA-ICC) located at the Environment Laboratories in Monaco. Since 2010, the United States has allocated a total of more than EUR 2 million (USD 2.3 million) to the OA-ICC.

The **United States** announced an Arctic Observing Network (US AON) initiative that will establish and support tasks to link existing U.S. agency contributions towards better integrated pan-Arctic observing networks in marine and terrestrial environments. Better integrated observing networks will enable access to high quality data, expertise, and information in

support of scientific understanding, stakeholder needs, and agency missions such as commerce, security, and sustainable marine ecosystems. This initiative directly supports the international Sustaining Arctic Observing Network (SAON) process, which aims to support and strengthen the development of multinational engagement for sustained and coordinated pan-Arctic observing and data sharing systems.

Aker BioMarine announced EUR 830,000 (USD 1 million) over the next five years to the Antarctic Wildlife Research Fund to further its work on promoting and facilitating research on the Antarctic ecosystem.

The **BNP Paribas Foundation** announced to support, with a donation of EUR 1.5 million, two research projects to enhance scientific knowledge on climate change impact on two major ecosystems: coral reefs and polar systems. For polar systems, the Foundation will support a consortium of 13 research teams from six countries to carry out the SENSEI project (SENtinels of the SEa Ice) that is to collect data on marine top predators as indicators of the sea ice ecosystem. For coral reefs, an international research team will carry out an exhaustive assessment of the role of each coral reef fish species in the different ecosystem services in the French Polynesian islands between 2017 and 2019. Moreover, by drawing on observations accumulated over more than thirty years in the Pacific, the team is to more accurately predict the consequences of current global warming on coral reefs.

The **Clean Arctic Alliance**, along with the European Climate Foundation, Eyak Preservation Council and the expedition cruise ship operator Hurtigruten announced their commitment to further develop the Arctic Commitment multistakeholder collaboration for a heavy fuel oil (HFO) free Arctic. The initiative will increase the groundswell of support for a ban on the use and carriage of HFO fuel in the Arctic. The aim is to expand the "Arctic Commitment" to over one hundred leading Arctic voices, including industry, environmental NGOs and indigenous communities; profile innovative technologies demonstrating that abandoning the use of HFO in the Arctic is feasible; profile innovative industry sector leaders using alternatives to HFO in the Arctic; launch new communications pieces on alternatives to HFO; and prepare an Arctic Commitment Report highlighting signatories, and "words of wisdom" from a selection of Arctic voices. The initiative will conclude towards the end of 2018.

Conservation International, The Nature Conservancy, International Union for Conservation of Nature, World Wildlife Fund and Wetlands International announced the launch of the Global Mangrove Alliance. The target of the Global Mangrove Alliance is to increase the global area of mangrove habitat 20 per cent over current extent by the year 2030, and an interim target to acquire pledges toward this goal from ten governments (with two-three from each region) and ten new partners (a mix of private sector and NGOs) by 2022. With an eye to catalysing new investments, the Alliance aims to improve land use management and on-the ground conservation, restoration and sustainable use of mangroves. The Alliance will work locally, regionally and globally to secure commitments from its partners, governments, and the private sector to halt and reverse mangrove loss.

Conservation International (CI) and Bechtel Corporation announced a new partnership to implement integrated "green-grey" infrastructure solutions for the most climate vulnerable communities across the developing world. Bringing together CI's 30 years of conservation

expertise with Bechtel, one of the most respected global engineering, construction, and project management companies, the partners will combine the power of natural ecosystems to deliver coastal protection, flood control, erosion and other critical services, with the surety of low impact traditional engineering infrastructure. Starting in the highly vulnerable Visayas region of the central Philippines, by 2018 the program will design "green-grey" solutions that simultaneously provide coastal protection from typhoons, sustainable livelihoods for coastal communities, and critical habitat for marine life in one of the most biodiverse regions of our ocean. The combination of 'green' nature-based solutions with 'grey' infrastructure will deliver solutions which maximise economic efficiency, minimise costs, and help vulnerable communities.

Coral Vita, a company that executes large-scale community-based restoration projects, reaffirmed its commitment to transplant 5,000 climate change resilient corals by 2020 from its Caribbean pilot coral farm. It announced that it will launch the first large-scale farm (~100,000 coral fragments) by 2021, using breakthrough methods to boost coral resiliency to climate change while accelerating growth rates up to 50x through its land-based farming models.

The **GEF** announced a total of EUR 17.8 million (USD 21 million) GEF investments to strengthen resilience and adaptive capacities in coastal areas vulnerable to climate change in Mozambique, Liberia, Guinea-Bissau and the Mediterranean Region.

The **International Alliance to Combat Ocean Acidification** announced that fifteen of its members will develop Ocean Acidification Action Plans by June 2019. The Alliance will also seek inclusion of ocean acidification and changing ocean conditions in the "nationally determined contribution plans" (NDC's) developed pursuant to the Paris Agreement at COP 23 and 24. The Alliance also announced its commitment to grow from its present 47 members to over 60 members by June 2018.

The **International Chamber of Shipping** announced to reduce CO₂ emissions to mitigate increases in ocean temperatures and prevent acidification by: (i) Maintaining international shipping's annual total CO₂ emissions below 2008 levels; (ii) Reducing CO₂ emissions per tonne-km by at least 20% by 2022, compared to 2008. Implementation will be achieved principally through technical and operational measures by shipping companies.

MAN Diesel & Turbo SE announced a facility of up to EUR 2 million allocated to 10 LNG-retrofits to improve environmental efficiency & footprint of existing vessels. MAN Diesel & Turbo SE promotes the use of liquefied natural gas (LNG) as replacement for heavy fuel oil and the transition to electricity-based synthetic fuels in order to achieve CO₂-neutrality by 2050.

The **Ocean Foundation** announced a EUR 1.05 million (USD 1.25 million) initiative for the ocean acidification capacity building for 2017 and 2018, particularly for developing nations, which will include workshops for policy and science capacity building as well as technology transfer for African, Pacific Island, Central American and Caribbean nations. This initiative, announced in 2016, has been expanded with regard to the increased funding commitments

from public and private partners, the number of scientists to be invited and number of kits to be gifted.

HAROPA, the alliance of ports of Le Havre, Rouen and Paris announced the commitment to improve its waste management and circular economy strategy by 2020, by ensuring the removal of all floating wastes from its ports and by establishing in the new fluvial Platform of the Port of Paris in Achères a sustainable logistic chain through waterway (instead of road). Being part of the Great Paris project, the chain aims to carry between 15 and 20 million cubic meter of building materials and building wastes.

Surfrider Foundation Europe announced it will work in partnership with ship-owners and key maritime stakeholders to reduce shipping emissions through the promotion of a label for sustainable shipping in France by 2025. This label will aim at certifying that businesses have measured and reduced their CO2 emissions during the transport chain of their products -from the producer to the seller- and making this effort visible to the consumers.

The **Science for Nature and People Partnership (SNAPP)** announced its commitment to deliver evidence-based, scalable solutions to global ocean challenges at the intersection of nature conservation, sustainable development, and human well-being. SNAPP working groups have explored science-informed solutions for a variety of ocean-related issues. There are 7 ocean-themed groups, involving 210 individuals, 76 institutions, and 23 countries: Coastal Defenses, Ridges to Reefs, Data Limited Fisheries, Fisheries Measures, Offshore Aquaculture, Coastal Restoration, and Assessing Biocultural Indicators. The research includes: efforts to value the coastal defenses from storms and sea level rise that are provided by nature and coastal preserves; development of a tool to help manage fisheries when there are limited data about the fisheries; determining how (and where) offshore aquaculture can be an integral component of global food security and sustainability; and how land-use decisions in Pacific islands impact the health of reef habitats and the management of local fisheries. The Partnership has invested EUR 1,1 million (USD 1,3 million) in this initiative and is committed to spend at least an additional EUR 500,735 (USD 598,537.42) on these projects."

The **World Economic Forum** and its members announced new additions to the list of companies, governments and international organisations who have full end to end traceability of tuna. They have also announced Global Ocean Data Alliance which comprises world's leading technology and data companies as well as scientific organisations. The Ocean Data Alliance will build technological tools needed for full end to end traceability of Tuna by 2020, in line with the Tuna Traceability Declaration.

XL Catlin announced its commitment to spend between EUR 2.1 and EUR 2.5 million (USD 2.5 - USD 3 million) over 2017-2018 on its Ocean Risk Initiative, which aims to help leaders to become better prepared to tackle potential consequences of ocean change and to generate new and dynamic solutions. In these two years, the initiative will: set up a new funding stream for Ocean Risk research by Early Career Scientists; organise the first Ocean Risk Summit (Bermuda, May 2018); deliver new risk reports and a global engagement program on the financial impacts of ocean warming, as well as continue funding the XL Catlin Oceans Education program, which aims to increase ocean literacy by reaching 10 million children by 2020.

Indonesia announced EUR 8.5 million to enhance the Indonesia Tsunami Early Warning System (InaTEWS) to provide inter alia meteorology, climatology, and geophysics service including public information, and early warning services.

Indonesia announced an EUR 850,000 commitment to conduct, in collaboration with the UN Economic and Social Commission for Asia and the Pacific (ESCAP), a project on Strengthening Multi-Hazard Risk Assessment and Early Warning Systems with Applications of Space and Geographic Information Systems in Pacific Island Countries.

Indonesia announced a commitment of EUR 850,000 to support a course programme to assist countries in the region in the area of disaster management. It will focus on enhancing the contribution and capability from charting in order to produce coastal and bathymetry mapping support for disaster assessment, disaster relief mapping and management, by its charting naval agency and hydrographic office in the region and other Archipelagic countries.

Indonesia announced it will restore the 18,882 square metres damaged coral reefs in the Dampier Strait off Raja Ampat (West Papua, Indonesia).

Indonesia announced the establishment of a national blue carbon program to support the establishment of the scientific, institutional and community engagement foundation necessary for Indonesia to benefit from future Blue Carbon finance opportunities, as well as alternative livelihood for coastal communities in its coastal mangrove and sea grass ecosystems. It will include the establishment of a Regional-International Center on Blue Carbon, expanding science-policy program, conservation and management, assessment of Blue Carbon value, mangrove & seagrass ecosystem restoration, capacity building in sustainable fisheries, tourism and alternative livelihoods.